From THESAN-ZOOM to JWST: Predicting ionizing photon escape and the rise of UV-bright reionization sources

Zebedee Summerfield [1,2,3★] William McClymont [2,3†] Sandro Tacchella [2,3‡] Aaron Smith [4] Rahul Kannan [5] Enrico Garaldi [6] Ewald Puchwein [7] Xuejian Shen [8] Josh Borrow [9] A. Lola Danhaive [2,3] Laura Keating [10] Gabriel Maheson [2,3,11,12] Parth Mehrotra [13] Charlotte Simmonds [14] Amanda Stoffers [2,3] Mark Vogelsberger [8] and Oliver Zier [15,8]

[1] Department of Physics, University of Oxford, Denys Wilkinson Building, Keble Road, Oxford OX1 3RH, UK
[2] Kavli Institute for Cosmology, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK
[3] Cavendish Laboratory, University of Cambridge, 19 JJ Thomson Avenue, Cambridge CB3 0HE, UK
[4] Department of Physics, The University of Texas at Dallas, Richardson, TX 75080, USA
[5] Department of Physics and Astronomy, York University, 4700 Keele Street, Toronto, ON M3J 1P3, Canada
[6] Kavli IPMU (WPI), UTIAS, The University of Tokyo, Kashiwa, Chiba 277-8583, Japan
[7] Leibniz-Institut für Astrophysik Potsdam, An der Sternwarte 16, 14482 Potsdam, Germany
[8] Department of Physics, Kavli Institute for Astrophysics and Space Research, Massachusetts Institute of Technology, Cambridge, MA 02139, USA
[9] Department of Physics and Astronomy, University of Pennsylvania, 209 South 33rd Street, Philadelphia, PA 19104, USA
[10] Institute for Astronomy, University of Edinburgh, Blackford Hill, Edinburgh, EH9 3HJ, UK
[11] Dipartimento di Fisica e Astronomia, Università di Bologna, Via Gobetti 93/2, I-40129, Bologna, Italy
[12] INAF, Astrophysics and Space Science Observatory Bologna, Via P. Gobetti 93/3, I-40129 Bologna, Italy
[13] Department of Physics, Indian Institute of Science Education and Research, Dr. Homi Bhabha Road, Pune, 411008, Maharashtra, India
[14] Departamento de Astronomía, Universidad de Chile, Camino El Observatorio 1515, Las Condes, Santiago, Chile
[15] Center for Astrophysics | Harvard & Smithsonian, 60 Garden Street, Cambridge, MA 02138, USA

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ABSTRACT

Understanding the sources and evolution of cosmic reionization remains a central challenge in astrophysics, with the escape of ionizing Lyman-continuum (LyC) photons from early galaxies representing a major uncertainty. In this work, we use more than 35,000 galaxy realisations from the THESAN-ZOOM cosmological radiation-hydrodynamic simulations to identify indirect diagnostics of the LyC photon escape fraction ($f_{\rm esc}$) and the LyC photon escape rate ($\dot{N}_{\rm ion,esc}$) across the redshift range $z = 3-16$. We train random forest regression models using these diagnostics to predict both quantities. We present four models: two trained with the full set of simulation-derived indicators to predict $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$, and two restricted to observables accessible to JWST photometric surveys. We find the 10-to-100 Myr star-formation rate ratio ($\rm SFR_{10}/SFR_{100}$) and the gas-to-stellar mass ratio ($M_{\rm gas}/M_*$) to be the strongest diagnostics of $f_{\rm esc}$, suggesting a strong relationship between ionizing photon escape and gas clearing through bursty star formation. In contrast, rest-frame UV (1500 Å) absolute magnitude ($M_{\rm UV}$) dominates $\dot{N}_{\rm ion,esc}$ prediction. Motivated by the strong predictive power of $M_{\rm UV}$, we combine observed UV luminosity functions with derived $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relations to construct histories of reionization. These are consistent with observational constraints, avoiding the recently reported crisis in the ionizing photon budget. Our analysis suggests that the bulk of reionization occurred rapidly after $z \approx 8$, driven by UV-bright galaxies, with the $M_{\rm UV} < -17$ populations providing the dominant contribution.

Key words: galaxies: evolution – galaxies: general – galaxies: high-redshift – dark ages, reionization, first stars

1 INTRODUCTION

The Epoch of Reionization (EoR) represents a major phase transition of the Universe, where the neutral hydrogen of the intergalactic medium (IGM) was ionized by the Lyman-continuum radiation (LyC; with energies in excess of 13.6 eV) emitted by the first stars and galaxies. While observational evidence indicates that reionization reached completion in the redshift range of $5 < z < 6$ (Becker et al. 2021; Bosman et al. 2022; Davies et al. 2024; Zhu et al. 2024), the onset and evolution of the transition remain uncertain.

Yet, in recent years, a wide variety of observational probes have enabled increasingly tight constraints. The early stages of reionization are indirectly constrained by Cosmic Microwave Background (CMB) measurements (Heinrich & Hu 2021), while recent surveys

★ E-mail: newc7894@ox.ac.uk (ZS)
† E-mail: wjm50@cam.ac.uk (WM)
‡ E-mail: st578@cam.ac.uk (ST)

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from the James Webb Space Telescope (JWST) have identified luminous galaxies at $z \geq 13$ exhibiting Lyman alpha (Lyα) emission (Carniani et al. 2024; Witstok et al. 2025), seemingly implying patchy reionization had already started at this early epoch. Additionally, the evolution of the hydrogen neutral fraction with cosmic time has been inferred from the decreasing abundance of Lyα emitters at higher redshift (Tang et al. 2024; Kageura et al. 2025), measurements of the Lyα and Lyβ forests (Jin et al. 2023; Zhu et al. 2024), and damping-wing analyses of high-redshift quasar spectra (Davies et al. 2018; Wang et al. 2020). However, there is still much debate regarding the nature of the sources responsible for reionization (Robertson 2022).

It is widely agreed that the ionizing photons produced by young, massive stars in star-forming galaxies play an important role in reionization (Madau 2017; Rosdahl et al. 2018; Yeh et al. 2023; Choustikov et al. 2024a). However, it remains unclear whether active galactic nuclei (AGN) make a significant contribution. JWST has revealed an emergent population of broad-line AGNs at $4 \lesssim z \lesssim 13$ (Maiolino et al. 2024; Fujimoto et al. 2024; Madau et al. 2024), and it is known that galaxies harbouring AGNs can leak a large fraction of their produced LyC radiation (Cristiani et al. 2016; Iwata et al. 2022; Madau et al. 2024). Although other studies argue that AGN must have a subdominant impact on reionization (Parsa et al. 2018; Puchwein et al. 2019; Trebitsch et al. 2020; Shen et al. 2020b), as their contribution is limited by number densities that only become relevant at $z \lesssim 7$ (Dayal et al. 2025), and by constraints on the thermal history of the IGM (Upton Sanderbeck et al. 2016; Puchwein et al. 2019). The community also remains divided over whether reionization is driven primarily by numerous faint galaxies or by the most luminous populations (Kakiichi et al. 2018; Naidu et al. 2020; Dayal et al. 2020; Larson et al. 2022; Atek et al. 2024; Mascia et al. 2024; Simmonds et al. 2024b).

The evolution of the EoR is characterised by the cosmic ionizing emissivity, $\dot{n}_{\rm ion}$, defined as the ionizing photons emitted per unit time per unit comoving volume. This quantity can be expressed as a function of redshift in terms of the mean galactic LyC photon escape rate, $\dot{N}_{\rm ion,esc}$, using:

$$\dot{n}_{\rm ion}(z) = \int_{M_{\rm UV,min}}^{M_{\rm UV,max}} \Phi(M_{\rm UV}, z)\ \left\langle \dot{N}_{\rm ion,esc} \,\middle|\, M_{\rm UV}, z \right\rangle \,{\rm d}M_{\rm UV}\,, \quad (1)$$

where Φ is the ultra-violet (UV) luminosity function, defined as the number density of galaxies as a function of redshift and rest-frame UV absolute magnitude. The integration limits $M_{\rm UV,min}$ and $M_{\rm UV,max}$ correspond to the faint- and bright-end magnitude limits of the galaxy population contributing to reionization, and are chosen on a case-by-case basis depending on observational constraints and modelling assumptions. Often, at the galactic level, $\dot{N}_{\rm ion,esc}$ is decomposed into:

$$\dot{N}_{\rm ion,esc} = \dot{N}_{\rm ion}\, f_{\rm esc} = L_{\rm UV}\, \xi_{\rm ion}\, f_{\rm esc} \quad (2)$$

for a galaxy of monochromatic UV luminosity $L_{\rm UV}$, where $f_{\rm esc}$ is the fraction of LyC photons that escape to the IGM and $\xi_{\rm ion}$ is the ionizing photon production efficiency. The latter is defined as the ratio of the ionizing photon production rate, $\dot{N}_{\rm ion}$, to the monochromatic UV luminosity, $L_{\rm UV}$: $\xi_{\rm ion} = \dot{N}_{\rm ion} \,/\, L_{\rm UV}$. In the absence of a direct estimate for $\dot{N}_{\rm ion,esc}$, which is unobservable during the EoR, all three quantities $L_{\rm UV}$ (or a UV luminosity density function $\rho_{\rm UV}$), $\xi_{\rm ion}$, and $f_{\rm esc}$ must be specified to study $\dot{n}_{\rm ion}$. Among these parameters, the escape fraction has historically been the most challenging to constrain, as it arises from complex, highly non-linear processes within galaxies, such as those governing the opacity of the interstellar medium (ISM) (Jaskot 2025). More importantly, the neutral IGM is opaque to ionizing radiation and thus direct measurements are not feasible at high redshifts, with no clear lines of sight above $z \sim 4$ (Inoue & Iwata 2008). Consequently, characterising $f_{\rm esc}$ remains a focus of active observational (Chisholm et al. 2022; Mascia et al. 2024; Jaskot et al. 2024; Mascia et al. 2025; Stoffers et al. 2025) and simulation-based research (Yoo et al. 2020; Rosdahl et al. 2022; Katz et al. 2022; Yeh et al. 2023; Kostyuk et al. 2023; Choustikov et al. 2024b,c).

Attempts to investigate LyC emitters observationally have been made through low-redshift analogues (Saxena et al. 2022b; Flury et al. 2022; Stoffers et al. 2025). However, it is unclear how representative these are of higher redshift reionization sources (Price et al. 2016; Katz et al. 2023b). Furthermore, recent JWST-based studies infer an enhanced abundance of UV-bright galaxies in the early Universe ($z \gtrsim 9$; Robertson et al. 2023, 2024; Carniani et al. 2024; Donnan et al. 2024; Tacchella 2024; Whitler et al. 2025); when combined with escape fractions inferred from low-redshift analogues (Chisholm et al. 2022), these results could imply an overproduction of ionizing photons during the EoR, leading to a "photon budget crisis" (Muñoz et al. 2024; Simmonds et al. 2024b).

Thus, due to the inability of observations to directly investigate ionizing photon emission during the EoR, theoretical models, and in particular numerical simulations, have become increasingly important. Fortunately, the fidelity of cosmological simulations has advanced significantly in recent years, driven by rapid developments in computational power and numerical methods (Vogelsberger et al. 2020). The study of reionization in a cosmological context requires the use of intermediate- or large-volume simulations that emulate regions spanning tens to hundreds of megaparsecs (Gnedin 2014; Rosdahl et al. 2018; Ocvirk et al. 2020; Kannan et al. 2022; Puchwein et al. 2023). While these capture the large-scale statistical process of reionization, the combination of limited resolution and simplified ISM and feedback physics hampers a detailed treatment of radiative transfer through the ISM, hindering robust predictions of $f_{\rm esc}$. Some studies instead employ idealised galactic simulations (Jensen et al. 2016; Yoo et al. 2020); however, their representativeness of LyC sources is ambiguous due to the absence of realistic feedback processes and a cosmological environment. Many recent works use a zoom-in simulation technique, where the resolution of a targeted halo is enhanced within a broader cosmological environment, enabling realistic modelling of radiation propagation (Ma et al. 2018; Pallottini et al. 2019; Rosdahl et al. 2022; Bhagwat et al. 2024; Choustikov et al. 2024c; Kannan et al. 2025). Indirect measurements of $f_{\rm esc}$ can therefore be obtained using models calibrated with these zoom-in mock galaxies (e.g., Choustikov et al. 2024b).

In this work, we develop predictive models, using the random forest (RF) regression machine learning technique for both the widely studied galactic LyC photon escape fraction, $f_{\rm esc}$, and the galactic LyC photon escape rate, $\dot{N}_{\rm ion,esc}$, which provides a more direct probe of a source's contribution to reionization. To do so, we employ the recently developed THESAN-ZOOM simulations (Kannan et al. 2025), a suite of zoom-in radiation-hydrodynamic simulations that resolve the multi-phase ISM with high spatial and mass resolution at $z > 3$. We apply our models to a sample of over 40,000 high-redshift JWST/NIRCam-detected JADES galaxies, deriving relations for $\dot{N}_{\rm ion,esc}$ dependent on redshift and rest-frame UV (1500 Å) absolute magnitude, $M_{\rm UV}$, for both the JWST sample and THESAN-ZOOM itself. Using these relations in Eq. (1), we infer implications for the EoR, finding a history of reionization in which the majority of the neutral hydrogen in the IGM is ionized rapidly after $z \sim 8$, driven predominantly by LyC emission from relatively UV-bright galaxies.

This paper is structured as follows. In Section 2, we introduce the THESAN-ZOOM simulations. We explore how potential diagnostics

are identified and how they are used to train RF predictive models in Section 3, where we also present and discuss our resulting best-performing and observationally applicable models. In Section 4, we explore the implications for reionization, calculating histories of $\dot{n}_{\rm ion}$, the hydrogen ionized fraction, $Q_{\rm HII}$, and the Thomson optical depth, τ, before presenting our conclusions in Section 5.

2 SIMULATION METHODOLOGY

2.1 The THESAN-ZOOM project

We employ the THESAN-ZOOM simulations Kannan et al. (2025), a state-of-the-art set of high-resolution zoom-in simulations that extends the original THESAN project (Kannan et al. 2022; Smith et al. 2022a; Garaldi et al. 2022, 2024).

The original THESAN project comprises a suite of large-volume ($L_{\rm box}$ = 95.5 cMpc) radiation-magneto-hydrodynamic simulations aimed at modelling the EoR and the early Universe, with galaxy formation following the IllustrisTNG model (Springel et al. 2018; Marinacci et al. 2018; Nelson et al. 2021). The THESAN simulations follow the prevailing ΛCDM cosmological framework and assume the cosmological parameters from the Planck Collaboration (Ade et al. 2016): $H_0 = 100\,h\,{\rm km\,s^{-1}Mpc^{-1}}$ with $h = 0.6774$, $\Omega_{\rm m} = 0.3089$, $\Omega_\Lambda = 0.6911$, $\Omega_{\rm b} = 0.0486$, $\sigma_8 = 0.8159$, and $n_{\rm s} = 0.9667$, where all symbols have the usual meaning. The simulations were initialised at $z_{\rm in} = 49$ (Kannan et al. 2022) with homogeneous expanding backgrounds and imposed Gaussian perturbations to instigate the formation of structure. THESAN was designed to capture the large-scale properties of the IGM and the galaxies driving reionization (e.g., Yeh et al. 2023; Borrow et al. 2023; Shen et al. 2024; Neyer et al. 2024; Zhao et al. 2025; Jamieson et al. 2025; Almualla et al. 2025), however, the effective equation-of-state treatment of the ISM limits THESAN's ability to model galactic processes, motivating the need for more advanced ISM and stellar feedback models to resolve escape at galactic scales.

Full details of THESAN-ZOOM are presented in Kannan et al. (2025), although we provide a brief overview here. THESAN-ZOOM uses a zoom-in technique to enhance the resolution and physics of 14 regions from the original THESAN simulations, enabling detailed studies probing galaxy-scale phenomena and properties during the EoR (McClymont et al. 2025a,b,c,d; Wang et al. 2025; Zier et al. 2025a,b; Pruto et al. 2026; Shen et al. 2026). The suite includes three distinct resolution tiers achieving enhancements in spatial resolution by factors of 4, 8, and 16 and corresponding increases in mass resolution by factors of 64, 512, and 4096 compared to the original THESAN runs. The corresponding baryonic mass resolutions are $9.09 \times 10^3\,{\rm M_\odot}$, $1.14\times10^3\,{\rm M_\odot}$ and $1.42\times10^2\,{\rm M_\odot}$, respectively (Kannan et al. 2025). THESAN-ZOOM inherits its initial conditions and cosmological parameters from THESAN, ensuring large-scale structures and cosmological properties are preserved.

The propagation of radiation is tracked across the boundaries of the zoom-in regions, using the time-varying radiation field from the parent volume as a boundary condition, to ensure consistency within the context of the large-scale environment (Kannan et al. 2025). The radiation-hydrodynamics solver AREPO-RT (Kannan et al. 2019), built on the AREPO moving mesh code (Springel 2010), and employing a novel node-to-node communication strategy (Zier et al. 2024), is used to self-consistently model LyC photon–gas interactions and radiative transfer "on-the-fly". It adopts a moment-based scheme with a reduced speed-of-light approximation (Kannan et al. 2019), accounting for both internal (zoom-in) and external radiation fields. This approach is crucial for accurately capturing the coupling between ionizing radiation and the ISM. A dust model is incorporated which tracks dust production, growth and destruction across five species (McKinnon et al. 2017; Kannan et al. 2020, 2025, Garaldi et al. in prep.). The stellar feedback combines multiple channels, including photoionization, radiation pressure, stellar winds, supernovae, and an additional early feedback channel (ESF) representing unresolved and unaccounted-for physics (Kannan et al. 2020, 2025). The ESF channel injects momentum into the gas surrounding newly formed star particles for the first 5 Myr after their formation, at a rate of $1000\,{\rm km\,s^{-1}\,Myr^{-1}}$ per unit stellar mass (Kannan et al. 2025). Its inclusion makes star formation less clustered and less bursty than in comparable simulation runs without ESF (Kannan et al. 2025). Stellar winds and star formation are implemented using the SMUGGLE multi-phase ISM model (Marinacci et al. 2019). These feedback processes regulate star formation and drive outflows, shaping early galaxy properties and facilitating radiation escape.

In this work, we limit our focus to simulations at the lowest resolution tier, with spatial resolution enhanced by a factor of four, ensuring consistent physical modelling across all galaxies and allowing us to include simulations which contain relatively massive galaxies at high redshift. A comparison with the available galaxy sample from the resolution tier enhanced by a factor of eight reveals qualitatively similar trends in LyC escape with UV luminosity, although systematic differences in their normalisation remain. We therefore acknowledge potential resolution dependence as an additional systematic uncertainty in the quantitative calibration of our inferred relations and their subsequent application to reionization. We consider only subhaloes resolved with at least 100 stellar particles, which corresponds to a stellar mass cut-off of $9.09 \times 10^5\,{\rm M_\odot}$. Haloes were identified using the friends-of-friends (FOF) algorithm (Davis et al. 1985), with self-gravitating haloes being identified within the FOF groups using the SUBFIND-HBT algorithm (Springel et al. 2001, 2021). Due to the computational expense of our analysis, we only consider the largest zoom region in the suite that was run down to $z = 3$, which is region m12.6 in Kannan et al. (2025). We include both central and satellite galaxies across $3 < z < 16$ in this analysis, which leaves us with a sample of 43 076 subhaloes. Stellar masses, star-formation rate, and luminosities are calculated based on particles within the virial radius. Gas masses are calculated within the galaxy (defined as within twice the UV half-light radius) in order to trace the gas within the ISM.

2.2 Post-processing of THESAN-ZOOM

The COsmic Lyman-α Transfer (COLT) code is a Monte Carlo Radiative Transfer (MCRT) solver for post-processing hydrodynamical simulations (Smith et al. 2015, 2019, 2022b; McClymont et al. 2025e, for public code access and documentation see colt.readthedocs.io). COLT can be run on simulation outputs to trace the paths of photons through the ISM and circumgalactic medium (CGM) (Tacchella et al. 2022; Smith et al. 2022b; McClymont et al. 2024, 2025a,b). The code treats radiation as individual photon packets, initialised with positions, directions and frequencies, that propagate through a 3D adaptive Voronoi tessellation. The photon packets are seeded using the BPASS v2.2.1 binary stellar population models (Eldridge et al. 2017; Stanway & Eldridge 2018), assuming a Chabrier (2003) initial mass function with an upper mass limit of 100 $M_\odot$, and interpolating the stellar spectra in age and metallicity for each star particle. These are the same stellar population models used to source the on-the-fly radiative transfer in THESAN-ZOOM (Kannan et al. 2025). COLT is run once per snapshot over the full high-resolution region, using approximately $1 - 2 \times 10^9$ photon packets per run. These are emitted from

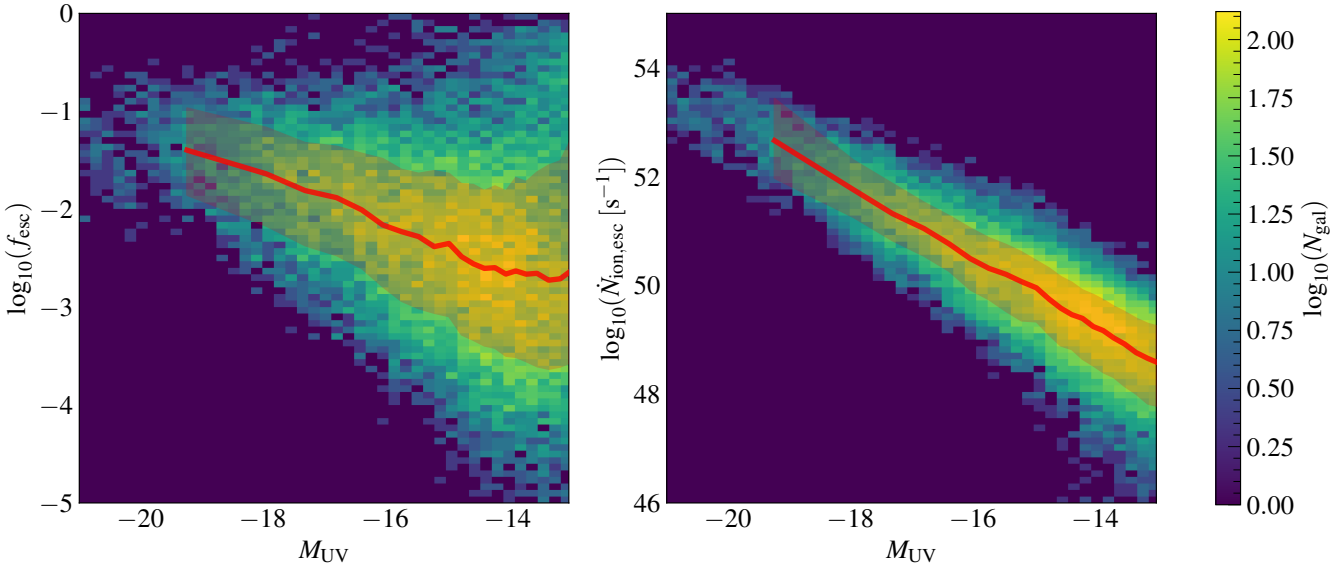


Figure 1. Ionizing photon escape fraction ($f_{\rm esc}$; *left panel*) and ionizing photon escape rate ($\dot{N}_{\rm ion,esc}$; *right panel*) as a function of rest-frame UV absolute magnitude ($M_{\rm UV}$) for the 35,512 galaxies selected from the THESAN-ZOOM simulations across $3 < z < 16$. The colour scale indicates the number of galaxies per bin. The red lines trace the median relations, with the red shaded regions denoting the $16^{\rm th} - 84^{\rm th}$ percentile range. The mean half-width of these intervals corresponds to 1σ scatters of 0.829 dex and 0.628 dex, respectively. The median trends show increasing escape fractions and escape rates towards brighter galaxies. However, the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation displays reduced scatter relative to that of $f_{\rm esc}$. Therefore, we expect $M_{\rm UV}$ to serve as a good indirect diagnostic for both these escape parameters, while providing a particularly strong tracer of $\dot{N}_{\rm ion,esc}$.

all star particles in proportion to their intrinsic luminosities, such that the number of photon packets associated with each galaxy scales with its intrinsic output. At each step, an optical depth for the packet is randomly selected, and the path length is calculated to determine whether it scatters or escapes. COLT models many physical processes: photoionization, recombination cascades, collisional excitation, and both absorption and scattering interactions with dust. COLT can therefore accurately model the anisotropic escape, scattering, absorption, and (re-)emission of ionizing, continuum, and line photons.

We have applied COLT to the THESAN-ZOOM simulations to generate a catalogue of galaxy realisations spanning $3 < z < 16$. LyC escape is considered based on photons which escape the virial radius, with $f_{\rm esc}$ representing the global escape fraction averaged over all angles. Both $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$ are integrated over the full hydrogen-ionizing spectrum (photon energies greater than 13.6 eV). COLT samples this spectrum using 31 frequency bins between 1 and 912 Å, with bin edges at the HI, HeI, and HeII ionization thresholds (912, 504 and 228 Å, respectively) and additional logarithmically spaced bins between them. Thus $f_{\rm esc} = \dot{N}_{\rm ion,esc}/\dot{N}_{\rm ion}$ is the photon-number-weighted escape fraction over the full ionizing spectrum, rather than a monochromatic escape fraction.

When calculating LyC escape and observable emission, we use the contribution of both the live dust model and an additional contribution of 40% of the metals to dust. This additional dust is added in each cell as a fixed fraction, $f_{\rm dust} = 0.4$, of the metal mass, on top of the live dust, and distributed between silicate and graphite grains in proportion to their live abundances. The total dust-to-gas ratio is therefore increased by $0.4Z$, where Z is the gas metallicity, while preserving the grain composition of the live dust model. Dust absorption and scattering in COLT are treated using the wavelength-dependent cross-sections, albedos, and scattering asymmetry parameters from the Milky Way $R_V = 3.1$ grain models of Weingartner & Draine (2001), applied separately to silicate, graphite, and PAH grains. One per cent of the carbonaceous dust mass is assigned to PAHs. Scattering is anisotropic, such that the resulting attenuation curve emerges self-consistently from the radiative transfer and depends on both the local grain composition and the relative geometry of stars and dust. The additional metal contribution to dust is motivated by the live dust model producing insufficient attenuation to reproduce the observed UV luminosity functions (UVLFs; Kannan et al. 2025). We verify that this dust contribution has a negligible impact on LyC escape, consistent with previous studies showing that LyC absorption by dust is insignificant due to an optical depth that is always much smaller than that of photoionization (Yoo et al. 2020; Mauerhofer et al. 2021; Rosdahl et al. 2022). $M_{\rm UV}$ is the observed rest-frame 1500 Å UV absolute magnitude calculated with COLT, excluding nebular continuum emission. However, we use the intrinsic value for the Hα luminosity, $L_{\rm H\alpha}$, assuming that a sufficiently accurate dust correction would be available observationally.

Most of a galaxy's luminosity is produced by massive O-type and B-type stars, which have lifetimes of less than 10 Myr (Sternberg et al. 2003). Consequently, high-redshift surveys struggle to identify galaxies with low star-formation rates, as these objects tend to be intrinsically faint. For consistency with such surveys, to which we aim for our models to be applicable, we exclude galaxies with no star formation over the preceding 50 Myr of their evolution ($\rm SFR_{50} = 0$). Following this selection, the catalogue contains 35 512 mock galaxy realisations.

3 LYC ESCAPE PREDICTION

3.1 Identifying potential diagnostics

We assess how LyC escape fractions and escape rates vary with galactic properties in the THESAN-ZOOM simulations. For an observable to

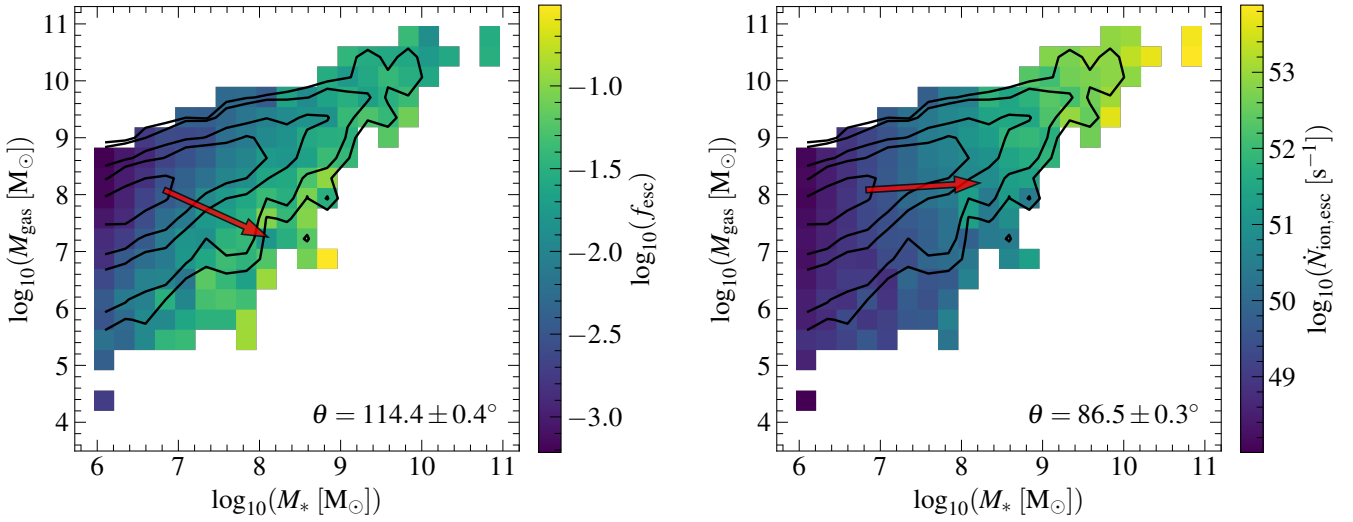


Figure 2. Gas mass ($M_{\rm gas}$) versus stellar mass (M_*) for THESAN-ZOOM galaxies, colour-coded by median ionizing photon escape fraction ($f_{\rm esc}$; *left-panel*) and median ionizing photon escape rate ($\dot{N}_{\rm ion,esc}$; *right panel*). Only bins containing at least five galaxies are plotted. The contours indicate the density of galaxies in the plots, with the innermost contours encompassing 34.0% of the sample and the outermost contours encompassing 97.5%. The red arrows indicate the direction of the strongest linear positive colour gradient, with their angles, θ, calculated from the partial correlation coefficients and defined clockwise from the positive y-direction. $\theta = 114.4 \pm 0.4° > 90°$ for the left plot, demonstrating that $f_{\rm esc}$ tends to increase as a galaxy's $M_{\rm gas}$ decreases and its M_* increases, implying that gas-to-stellar mass ratio is likely a good indicator for $f_{\rm esc}$. Conversely, $\theta = 86.5 \pm 0.3° < 90°$ for the right plot which indicates that both $M_{\rm gas}$ and M_* are positively correlated with $\dot{N}_{\rm ion,esc}$, with $M_{\rm gas}$ exhibiting a relatively weak correlation, which suggests this ratio is not an important diagnostic of $\dot{N}_{\rm ion,esc}$. This highlights how $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$ are distinct quantities governed by different physical processes.

serve as an indirect diagnostic, the median $f_{\rm esc}$ must exhibit a clear systematic trend with that property; the same applies to diagnostics of $\dot{N}_{\rm ion,esc}$.

In the left panel of Fig. 1 we consider the variation of $f_{\rm esc}$ with $M_{\rm UV}$. The red line of median $f_{\rm esc}$ against median $M_{\rm UV}$ shows a predominantly monotonic-decreasing relationship down to $M_{\rm UV} \approx -13$, implying high UV luminosity is ascociated with a higher median escape fraction for THESAN-ZOOM galaxies in the range $-20 < M_{\rm UV} < -13$. We see that faint galaxies can still achieve very high escape fractions; however, their prominance reflects the increased scatter and larger sample size at lower luminosities, rather than the typical trend which is more relevant for our predictive models. Furthermore, resolution effects in THESAN-ZOOM become increasingly relevant towards the faint end, where we impose the stellar-mass completeness limit of $M_* \sim 10^6 {\rm M}_\odot$. These effects should be kept in mind when interpreting Fig. 1 around $M_{\rm UV} \approx -13$.

The median relationship we find between $f_{\rm esc}$ and $M_{\rm UV}$ differs from contemporary results, warranting further discussion. Most observational studies into this relationship find either an anti-correlation between $f_{\rm esc}$ and $M_{\rm UV}$ (Chisholm et al. 2022; Saxena et al. 2022a; Pahl et al. 2023), or no discernible trend at all (Giovinazzo et al. 2026; Jecmen et al. 2026). However, the trend found in the Low-redshift Lyman Continuum Survey by Chisholm et al. (2022) may not be representative of the high-redshift population considered here. Similar considerations apply to the surveys used by Saxena et al. (2022a); Pahl et al. (2023), which probe galaxies at $z \sim 3$, well after the EoR. Previous simulation results are in less conflict with ours. Naidu et al. (2020) find that $f_{\rm esc}$ grows with stellar mass, which is strongly correlated with $M_{\rm UV}$, using an empirical model that successfully reproduces observations of the EoR. Yeh et al. (2023) employ the THESAN simulations and conclude that there is no simple relationship between $f_{\rm esc}$ and star formation rate (SFR; also well-traced by $M_{\rm UV}$), although they do find that galaxies with higher SFR tend to have higher $f_{\rm esc}$. However, THESAN adopts different galaxy-formation models from THESAN-ZOOM, while the explicit modelling of the multiphase ISM in THESAN-ZOOM enables a more detailed treatment of LyC escape (Marinacci et al. 2019; Nelson et al. 2021; Kannan et al. 2025). The SPHINX Simulations provide a closer comparison (Rosdahl et al. 2018), with Rosdahl et al. (2022) demonstrating that SPHINX galaxies exhibit increasing $f_{\rm esc}$ towards brighter galaxies, broadly similar to our results, but with a turnover at the bright end. This difference at high luminosities may be related to the tendency of higher-mass SPHINX galaxies to form more ordered disks with smoother star-formation histories when compared to THESAN-ZOOM galaxies which tend to remain more disordered and bursty, potentially facilitating more efficient LyC escape (McClymont et al. 2025b,a; Shen et al. 2026).

We also consider the relationship of $\dot{N}_{\rm ion,esc}$ with $M_{\rm UV}$ in the right panel of Fig. 1, similarly finding that escape rate increases with luminosity over the same range. Comparing the plots in Fig. 1, the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation appears stronger, exhibiting a strictly monotonic trend. This is further reflected in its smaller scatter: the mean half-width of the $16^{\rm th} - 84^{\rm th}$ percentile range (the 1σ scatter) is 0.628 dex for $\dot{N}_{\rm ion,esc}$ as a function of $M_{\rm UV}$, compared with 0.829 dex for $f_{\rm esc}$. Moreover, the increase in scatter towards the faint end is far less pronounced than for $f_{\rm esc}$. Thus, the median $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation is tighter and exhibits reduced variance relative to its dynamic range. We therefore expect $M_{\rm UV}$ to be a useful diagnostic of both $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$, but a substantially stronger tracer of $\dot{N}_{\rm ion,esc}$. To verify that the difference in scatter is not driven by redshift evolution, we subdivide the galaxy sample into three redshift bins and find minimal variation in both median relations and their scatter across the bins (not shown in Fig. 1 to avoid overcrowding). This implies that the observed variance is driven primarily by intrinsic galaxy-to-galaxy stochasticity.

We also inspect ratios between different galactic properties in the

search for potential indicators. In Fig. 2, we plot the THESAN-ZOOM galaxies in 2D histograms of gas mass $M_{\rm gas}$ against stellar mass M_*, where the colour scales encode $f_{\rm esc}$ (left panel) and $\dot{N}_{\rm ion,esc}$ (right panel). The red arrows mark the direction of the steepest positive linear colour gradient on the 3D surface of each diagram, determined via a Partial Correlation Coefficient analysis (Bluck et al. 2020), adapted from Maheson et al. (2024). The corresponding angles θ are defined clockwise from the positive y-direction. In the left panel, the arrow lies at an angle of $\theta = 114.4 \pm 0.4° > 90°$. Thus, $f_{\rm esc}$ tends to increase as a galaxy's $M_{\rm gas}$ decreases and its M_* increases, suggesting gas-to-stellar mass ratio could be a good indicator for $f_{\rm esc}$. Conversely, the arrow in the right panel lies at an angle of $\theta = 86.5 \pm 0.3° < 90°$, indicating $M_{\rm gas}$ and M_* are both positively correlated with $\dot{N}_{\rm ion,esc}$. The near-orthogonality of this angle further conveys that $\dot{N}_{\rm ion,esc}$ exhibits a relatively weak relationship with $M_{\rm gas}$ compared with M_*. This suggests that gas-to-stellar mass ratio is likely to be a far less important diagnostic of $\dot{N}_{\rm ion,esc}$ than of $f_{\rm esc}$, highlighting that $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$, while related, are physically distinct quantities governed by different processes and constraints. It is therefore unsurprising that they exhibit different sets of indicators.

In the search for potential indicators of $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$, we perform the same single-variable analysis as done for $M_{\rm UV}$, applied to many parameters in the THESAN-ZOOM catalogue suspected to have predictive power. Analogous plots for galactic properties successfully identified as indicators are shown in Appendix A (see Figs. A1 and A2). We also repeat the ratio analysis using 2D histograms, as in Fig. 2, for multiple physically motivated pairs of parameters from the catalogue (see Fig. A3 in Appendix A). This yields a set of potential diagnostics, which, after further filtering explained in the following subsection, can be seen in Table 1.

Indicator	Definition	Models
$\rm SFR_{10}$	star-formation rate over 10 Myr	B, D
$\rm SFR_{100}$	star-formation rate over 100 Myr	B, D
$\Delta\rm MS_{10}$	Offset from the star-forming main sequence over 10 Myr; formula in Eq. (3)	A, C
$\rm SFR_{10}/SFR_{100}$	10–to–100 Myr star-formation rate ratio	A, C
M_*	Stellar mass	A, B, C, D
$M_{\rm gas}/M_*$	Ratio of gas mass to stellar mass	A, B
$M_*/M_{\rm vir}$	Ratio of stellar mass to halo virial mass	A, B
$R_{\rm SFR}$	Half-radius of star-formation rate	A
$R_{\rm SFR}/R_{M_*}$	Ratio of half-radius of star-formation rate and stellar half-mass radius	A
$\Sigma_{\rm SFR_{10}}$	star-formation rate surface density over 10 Myr; formula given in Eq. (5)	A
Z	Mean stellar metallicity	A, B
$M_{\rm UV}$	Rest-frame 1500 Å Observed UV absolute magnitude	A, B, C, D
$A_{\rm UV}$	Rest-frame 1500 Å UV Attenuation	A, B
$L_{\rm H\alpha}$	6562.8 Å Hα line integrated luminosity	B
$L_{\rm UV}/L_{\rm H\alpha}$	Ratio of UV (1500 Å) to Hα luminosities	A
$R_{\rm UV}$	Half-light radius of 1500 Å UV luminosity	A
$R_{\rm H\alpha}$	Half-light radius of Hα luminosity	A

Table 1. The indirect diagnostics, their definitions and which of our $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$ prediction models use them in training. All quantities are calculated based on particles within the virial radius, except for gas masses, which are calculated within the galaxy (defined as within twice the UV half-light radius) in order to trace the gas within the ISM. Model A: our best $f_{\rm esc}$ predictor trained with all the relevant THESAN-ZOOM-derived diagnostics; Model B: similarly our best $\dot{N}_{\rm ion,esc}$ predictor; Model C: our $f_{\rm esc}$ predictor trained only with features compatible with photometric JWST surveys; Model D: our photometrically compatible $\dot{N}_{\rm ion,esc}$ predictor.

3.2 Random forests

We build our prediction models using random forest (RF) regression (Breiman 2001). This is a machine learning method that constructs decision trees during training and outputs their mean prediction upon application. The investigation dataset, comprising features and the target variable for prediction, is split into a training set and a testing set, with the test sample used to evaluate the accuracy of the regressor. The error is typically quantified by Mean Squared Error (MSE) and Mean Absolute Error (MAE) measurements, which measure the average discrepancy between the random forest's predictions and the dataset values. Each tree in the forest is trained on a bootstrap sample of the training data, generated by sampling with replacement, which introduces diversity among the trees. At each split, a random subset of features is considered, and node decisions are chosen to minimise the error criterion, chosen to be MSE here. The randomness of bootstrapping and feature selection enables the RF regressor to capture complex, nonlinear relationships in the data while remaining robust to noise and outliers. After training, the feature importances of the model are returned, which represent the average contribution of each feature to the decrease in the error criterion. The importances in this work, given as fractions that sum to 1, represent how dependent the RF is on each observable for the prediction of the target variables, which are $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$ here. The random forest technique has been widely applied in recent astrophysical studies, demonstrating its versatility and effectiveness across a range of problems (Bluck et al. 2022; Baker et al. 2023; Maheson et al. 2024; Kawinwanichakij et al. 2026).

We used the RF regressor from the PYTHON package SCIKIT-LEARN for this analysis (Pedregosa et al. 2011). In each training sequence, the data from the THESAN-ZOOM catalogue was first divided into a train and test sample, with a 75:25 split. We use subsets of the identified observable indicators as the features of the dataset, applying base-10 logarithms to each (excluding those already in magnitudes). In all models, we include $\log_{10}(1+z)$ to track cosmic evolution and a uniform random variable for benchmarking. To avoid log(0) infinities, we add a small ϵ_x to every value of feature x, defined as $\epsilon_x = 0.01 \min\{x\}$, where $\min\{x\}$ denotes the smallest non-zero instance of feature x. To enhance the regressors' accuracy and minimise overfitting, we fine-tune specific execution parameters within the RF function, referred to as hyperparameters; Appendix B explains this optimisation.

To finalise the indicator sets, we trained 100 preliminary models for each of the targets $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$, using the previously identified potential indicators to obtain the mean feature importances. Regression was then applied to each of the 100 test samples to compute the mean prediction MAE. For any feature with an importance below 0.01, a further 100 models were trained without it; if the removal of that feature resulted in a lower mean test MAE, it was classified as irrelevant and removed from the potential indicator set. This iterative procedure ultimately yielded 14 indicators for $f_{\rm esc}$ and 9 for $\dot{N}_{\rm ion,esc}$ which we present in Tab. 1.

For completeness, we provide below the expressions for several derived parameters listed in Tab. 1. The specific star-formation rate, $\rm sSFR_\tau$, for a galaxy with stellar mass M_* and star-formation rate $\rm SFR_\tau$ over a lookback time of τ in Myr, is given by $\rm sSFR_\tau = SFR_\tau/M_*$. The majority of star-forming galaxies follow a relatively tight relation between sSFR and M_* known as the "main sequence" (Speagle et al. 2014; Tacchella et al. 2016; Popesso et al. 2023; Clarke et al. 2024; McClymont et al. 2025b; Simmonds et al. 2025). Offset

Model	Target	MAE	MSE	RMSE	PCC
A	$f_{\rm esc}$	0.420	0.305	0.552	0.809
B	$\dot{N}_{\rm ion,esc}$	0.407	0.289	0.537	0.919
C	$f_{\rm esc}$	0.476	0.386	0.621	0.739
D	$\dot{N}_{\rm ion,esc}$	0.446	0.351	0.592	0.899

Table 2. The mean absolute error (MAE; in dex), mean squared error (MSE; in dex^2), root mean squared error (RMSE; in dex), and Pearson Correlation Coefficient (PCC; ranging from -1 to 1), for our models calculated from their respective test samples. Model A: our best $f_{\rm esc}$ predictor trained with all the relevant THESAN-ZOOM-derived diagnostics. Model B: similarly our best $\dot{N}_{\rm ion,esc}$ predictor. Model C: our $f_{\rm esc}$ predictor trained only with features compatible with the JWST observational sample from Simmonds et al. (2025). Model D: our observationally compatible $\dot{N}_{\rm ion,esc}$ predictor.

from the star-forming main sequence over τ Myr, $\Delta\mathrm{MS}_\tau$, is defined at z for a galaxy with stellar mass M_* as:

$$\Delta\mathrm{MS}_\tau = \log_{10}\left(\frac{\mathrm{sSFR}_\tau}{\mathrm{sSFR}_{\rm MS}(M_*, z)}\right), \tag{3}$$

where the main sequence specific star-formation rate has the functional form (Tacchella et al. 2016):

$$\mathrm{sSFR}_{\rm MS}(M_*, z) = s_b \left(\frac{M_*}{10^{10}\,\mathrm{M}_\odot}\right)^{\beta} (1+z)^{\mu} \quad \mathrm{Gyr}^{-1}. \tag{4}$$

For a look back time of $\tau = 10$ Myr, we use the best fit parameters $s_b = 0.033\pm0.002, \beta = 0.041\pm0.004$ and $\mu = 2.64\pm0.03$, calculated by McClymont et al. (2025b). We also define the SFR surface density, $\Sigma_{\mathrm{SFR}_{10}}$, using the half-radius of SFR, $R_{\rm SFR}$:

$$\Sigma_{\mathrm{SFR}_{10}} = \frac{\mathrm{SFR}_{10}\,/\,2}{\pi\,(R_{\rm SFR})^2}. \tag{5}$$

Finally, the UV Attenuation is defined as the difference between the observed 1500 Å UV magnitude and the intrinsic (unobscured by absorption and scattering) 1500 Å UV magnitude of a galaxy: $A_{\rm UV} = M_{\rm UV,obs} - M_{\rm UV,int}$.

The features of our models are limited to observables obtainable through COLT post-processing. In addition to the diagnostics identified here, the UV spectral slope, β, is often considered a useful tracer of $f_{\rm esc}$ and is incorporated in the models of Chisholm et al. (2022), Choustikov et al. (2024b), and Stoffers et al. (2025). β is strongly correlated with UV dust attenuation (Shen et al. 2020a; Jecmen et al. 2026); it also encodes information about the stellar populations and the state of the ISM (Saxena et al. 2026). Since our framework includes UV attenuation together with several star formation diagnostics, we expect part of the information content of β to be captured indirectly. Nevertheless, we acknowledge that explicitly including β, as well as other tracers proposed in recent studies (Katz et al. 2022; Xu et al. 2022; Rosdahl et al. 2022; Katz et al. 2023a; Mascia et al. 2023, 2024; Choustikov et al. 2024c; Jaskot et al. 2024), could provide further predictive power.

3.3 Model results

We present four models trained using the THESAN-ZOOM catalogue with the indicator sets specified in Tab. 1. For each indicator configuration, 1000 random forests were trained, and the regressor yielding the smallest mean absolute error (MAE; in dex) on prediction for its corresponding test sample was selected as the final model. Their MAE, mean squared error (MSE; in dex^2), root mean squared error (RMSE; in dex), and Pearson correlation coefficient (PCC), which quantifies the strength of the linear relationship between the true and predicted values and ranges from -1 to 1, are reported in Tab. 2. Model A is our best $f_{\rm esc}$ predictor, trained using all 14 of the finalised $f_{\rm esc}$ indicators. Model B is our best $\dot{N}_{\rm ion,esc}$ predictor, similarly trained using all 9 of the identified $\dot{N}_{\rm ion,esc}$ indicators.

Models C and D likewise predict $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$, respectively, but are trained using only observables that are typically available in JWST photometric catalogues. Their indicator sets are therefore subsets of those of Models A and B, reflecting the more limited variable coverage of photometric catalogues relative to THESAN-ZOOM. Consequently, Models C and D perform marginally worse than A and B, respectively, across all error and correlation metrics shown in Tab. 2.

The feature importances of the models are presented in Fig. 3 as the mean importances averaged over their respective 1000 RF train-test realisations. We find the 10–to–100 Myr star-formation rate ratio, $\mathrm{SFR}_{10}/\mathrm{SFR}_{100}$, to be the best diagnostic of $f_{\rm esc}$, followed closely by the gas-to-stellar mass ratio, $M_{\rm gas}/M_*$, and then rest-frame UV (1500 Å) absolute magnitude, $M_{\rm UV}$. Conversely, $M_{\rm UV}$ is found to dominate predictions of $\dot{N}_{\rm ion,esc}$, with significant importance also placed on the star-formation rate observables SFR_{100} and SFR_{10}. The uniform random variable, included only for benchmarking, is reassuringly the least important predictive feature for each model.

The predictions of the models are plotted against the true $f_{\rm esc}$ or $\dot{N}_{\rm ion,esc}$ values of their respective test samples in Fig. 4. The prediction errors for all these models tend to decrease for higher predicted values, as demonstrated in the lower panels of Fig. 4, which is encouraging given that high-escape galaxies are the most relevant for reionization. We see a larger scatter around the 1 : 1 line for the $f_{\rm esc}$ models when compared to the $\dot{N}_{\rm ion,esc}$ models, which results in higher values of the PCC for the $\dot{N}_{\rm ion,esc}$ predictors, as seen in Tab. 2. This implies our models constrain $\dot{N}_{\rm ion,esc}$ more effectively, suggesting that $\dot{N}_{\rm ion,esc}$ may therefore be a more reliable quantity for reionization modelling.

A further limitation of our $f_{\rm esc}$ models is their inability to produce predictions at the highest escape fractions, particularly in the range $\log_{10}(f_{\rm esc}) > -0.5$ (see the left two panels of Fig. 4). This reflects the inherent nature of random forests, which average over decision tree outputs, thereby smoothing extreme values and restricting predictions to the range spanned by the THESAN-ZOOM training data. However, this limitation affects only a small subset of galaxies, and the resulting upper bound on predicted values lies approximately within one MAE of the physical maximum $\log_{10}(f_{\rm esc}) = 0$, limiting its practical impact. While our $\dot{N}_{\rm ion,esc}$ predictors exhibit a similar, albeit softer, capping behaviour, this quantity has no intrinsic upper limit, rendering the effect largely inconsequential.

We test Models C and D using the photometric catalogue presented in Simmonds et al. (2025), which is based on JWST NIRCam imaging of the JWST Advanced Deep Extragalactic Survey (JADES) in the GOODS-South and GOODS-North fields (Eisenstein et al. 2023; Bunker et al. 2024). The catalogue includes stellar masses, star-formation rate, and rest-frame 1500 Å UV magnitudes inferred via SED fitting with PROSPECTOR (Johnson et al. 2021), adopting a Chabrier initial mass function (IMF) (Chabrier 2003), a two-component dust model (Charlot & Fall 2000; Conroy et al. 2009), and non-parametric star-formation histories (SFHs) (Leja et al. 2019), where SFH is described with eight different SFR bins. Photometric redshifts were obtained using the fitting code EAzY (Brammer et al. 2008) and used as priors in the SED fitting. The following selection criteria were imposed on the JADES photometry dataset: sources must have a signal-to-noise ratio SNR$\geq$ 3 in the F444W NIRCam band, photometric redshift in the range $3 \leq z \leq 9$; and a

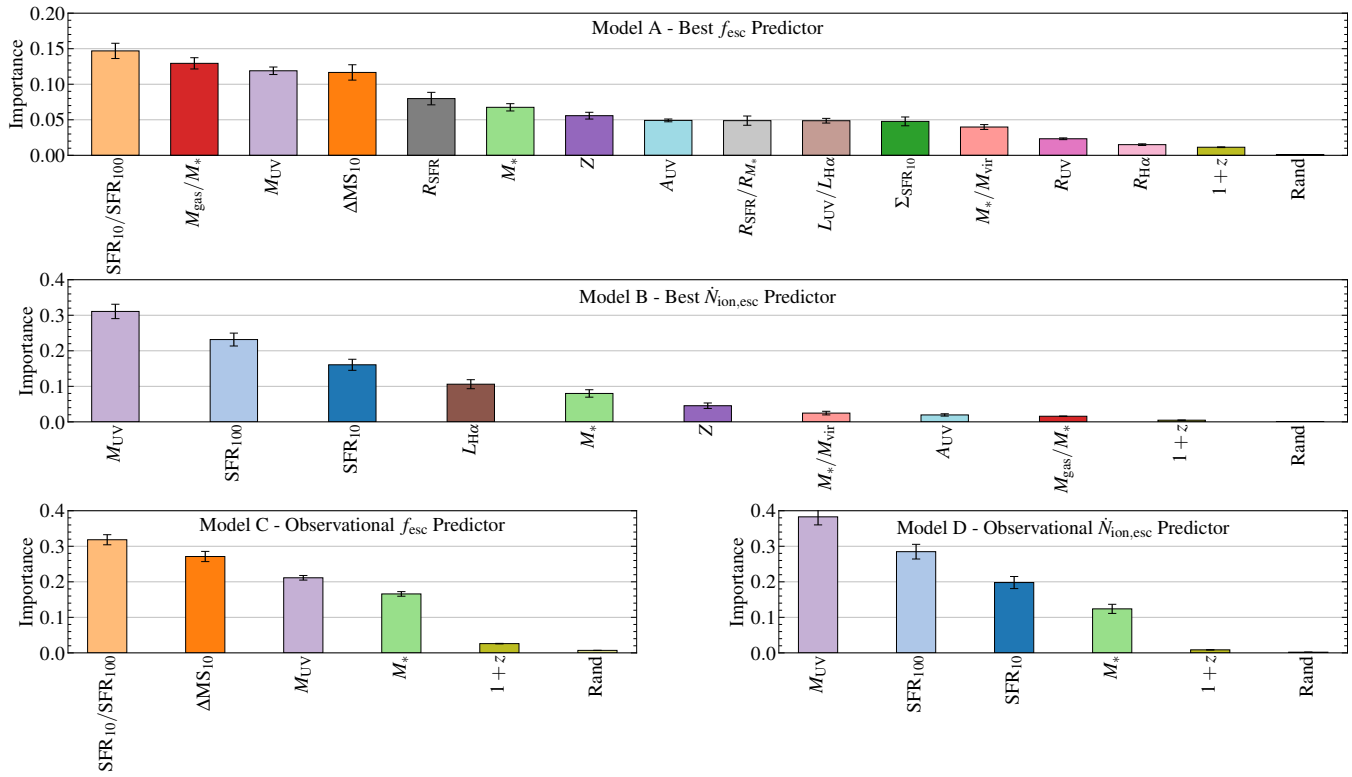


Figure 3. Mean feature importances for the four random forest models, averaged over 1000 train–test realisations. Importances represent the fractional contribution of each feature to reducing the mean squared error. The error bars on each importance denote the respective standard deviation across the 1000 runs. *Top panel*: Model A, our best $f_{\rm esc}$ predictor trained with all the identified THESAN-ZOOM $f_{\rm esc}$ indicators. *Middle panel*: Model B, similarly our best $\dot{N}_{\rm ion,esc}$ predictor. *Bottom left panel*: Model C, our $f_{\rm esc}$ predictor trained using only features compatible with photometric JWST observations. *Bottom right panel*: Model D, our photometrically compatible $\dot{N}_{\rm ion,esc}$ predictor. The indicators are defined in Tab. 1. The uniform random variable is reassuringly the least important predictive feature for each model relative to the chosen model indicators. This ranking reveals that the ratios $\mathrm{SFR_{10}/SFR_{100}}$ and $M_{\rm gas}/M_*$ are the strongest diagnostics of $f_{\rm esc}$, whereas the rest-frame UV absolute magnitude, $M_{\rm UV}$, clearly dominates predictions of $\dot{N}_{\rm ion,esc}$.

reduced $\chi^2 \leq 10$. The resulting catalogue contains 41,430 galaxies. 99.7% of these galaxies fall within $-21.7 < M_{\rm UV} < -13.1$ and $6.2 < \log_{10}(M_*/\mathrm{M}_\odot) < 10.5$ (the 3σ observed UV magnitude and stellar mass ranges).

Our predictions of $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$, using models C and D respectively for the JWST photometric sample from Simmonds et al. (2025), are plotted in Fig. 5 against $M_{\rm UV}$ and M_*. In both lower panels, $\log_{10}(\dot{N}_{\rm ion,esc})$ shows an approximately linear trend with $M_{\rm UV}$ and $\log_{10}(M_*)$. We therefore use orthogonal distance regression to fit linear relationships to each (Hughes 2010), with the resulting fit parameters presented in their respective panels. We include redshift dependence in both to account for the cosmic evolution of $\dot{N}_{\rm ion,esc}$. For comparison, the analogous $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$ plots and fits for the intrinsic THESAN-ZOOM values are shown in Fig. A4 in Appendix A. The $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation obtained from this JWST analysis closely matches that found in THESAN-ZOOM, albeit with an even weaker redshift dependence. A similarly weak redshift dependence of $\dot{n}_{\rm ion}$ inferred from JWST/NIRCam photometry has also been reported by Choustikov et al. (2024a). Our JWST-based relation also exhibits very little scatter, implying that the model relies most strongly on $M_{\rm UV}$ in prediction, in agreement with the feature importances shown in Fig. 3.

3.4 Indirect indicator physics

3.4.1 $f_{\rm esc}$ *indicator physics*

Massive O-type and B-type stars, with lifetimes of less than 10 Myr (Sternberg et al. 2003), are responsible for the majority of LyC photon emission. Furthermore, their winds and supernovae create the low-density channels through which LyC photons can escape (Sharma et al. 2017). Consequently, large populations of these short-lived stars are linked to high $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$ (Choustikov et al. 2024b; Bhagwat et al. 2024), in agreement with the high importance our models place on the star-formation rate observables $\Delta\mathrm{MS_{10}}$, $\mathrm{SFR_{10}}$, and $\mathrm{SFR_{100}}$ (see Fig. 3).

Many studies argue that periods of rapid, bursty star formation generate the intense stellar feedback that drives large LyC escape (Heckman et al. 2011; Katz et al. 2023a; Bhagwat et al. 2024; Jaskot 2025). In particular, Bhagwat et al. (2024) find that galaxies with high $\mathrm{SFR_{10}/SFR_{100}}$, indicative of ongoing starbursts, tend to exhibit high $f_{\rm esc}$. Our results likewise demonstrate that this ratio is the most important parameter governing $f_{\rm esc}$ (see Fig. 3); however, in THESAN-ZOOM we find the opposite behaviour: $\mathrm{SFR_{10}/SFR_{100}}$ has a strict negative correlation with $f_{\rm esc}$ (see Fig. A1), where $f_{\rm esc}$ is highest when $\mathrm{SFR_{10}/SFR_{100}} \lesssim 0.1$, corresponding to a post-burst or temporarily quenched phase. Physically, this can be understood as dense gas surrounding young massive stars during the burst absorbing most ionizing photons, whereas subsequent feedback

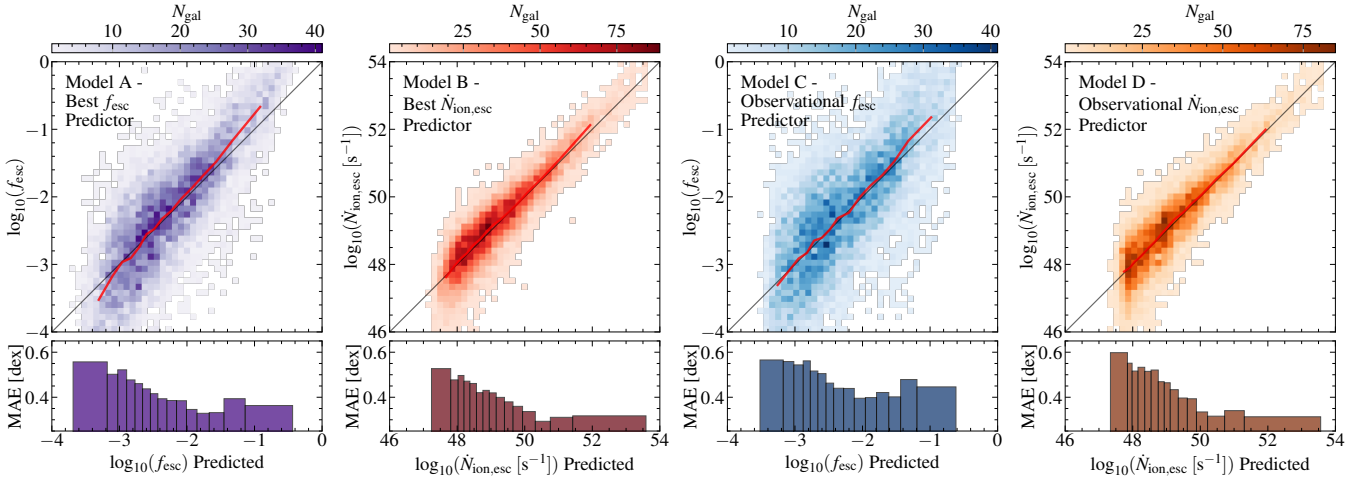


Figure 4. True versus predicted values for the four random forest models presented. From left to right they are: Model A, our best f_{esc} predictor trained with all the identified THESAN-ZOOM f_{esc} indicators; Model B, similarly our best $\dot{N}_{ion,esc}$ predictor; Model C, our f_{esc} predictor trained using only features compatible with photometric JWST surveys; Model D, our photometrically compatible $\dot{N}_{ion,esc}$ predictor. Top panels show predicted against true $\log_{10}(f_{esc})$ or $\log_{10}(\dot{N}_{ion,esc})$, with the colour scale indicating the number of galaxies per bin. Red curves trace the median relation. Bottom panels show the mean absolute error (MAE; in dex) as a function of predicted value. All models achieve higher accuracy when predicting larger values of f_{esc} and $\dot{N}_{ion,esc}$. Moreover, predictions of $\dot{N}_{ion,esc}$ exhibit systematically reduced scatter and higher accuracy, highlighting their suitability for reionization modelling.

clears the ISM and opens the low-density channels that enable delayed LyC escape. This interpretation is consistent with Katz et al. (2023a), who identify a population of "Remnant Leakers" with low SFR_{10}/SFR_{100} and high f_{esc} in the SPHINX^{20} simulation (Rosdahl et al. 2022), where supernova-driven feedback from previous bursts has disrupted the ISM and facilitated leakage. Similarly, the strongest leakers in THESAN-ZOOM lie below the star-forming main sequence, with f_{esc} decreasing by approximately an order of magnitude from $\Delta MS_{10} \sim -1$ to 0 (see Fig. A1), further supporting the association of strong LyC escape with a post-burst phase. This remnant-leaker behaviour persists despite the early stellar feedback implementation in THESAN-ZOOM, with the strongest LyC escape occurring after the 5 Myr ESF phase associated with a preceding burst. More generally, We have explored the inclusion of additional SFR timescales but found them to be insignificant in our RF models due to large degeneracy with the information already captured by SFR_{10}/SFR_{100}, hence their omission. We have similarly neglected any ionizing-luminosity-weighted stellar age tracers based on the same degeneracy argument.

Rosdahl et al. (2022) claim that smaller mass galaxies have less sustained starbursts, meaning their feedback is less disruptive and weaker in facilitating ionizing photon escape, supporting M_* as a key indicator of f_{esc}, consistent with our THESAN-ZOOM findings. M_{UV} is similarly important to our f_{esc} models due to its strong association with M_* and star-formation rates, reflecting its dependence on the number of young, massive stars. Likewise, the well-established correlation between M_* and UV dust attenuation supports the relevance we find for A_{UV} in predicting f_{esc} (Pannella et al. 2015; Shen et al. 2020a).

Morphological variables seem to play only a limited role in f_{esc} regression and have negligible importance in $\dot{N}_{ion,esc}$ prediction (see Fig. 3). We note that these results highlight the relative strength of different predictors. Additionally, where there are variables with degenerate predictive power, this method generally assigns importance to the "better", or more fundamental, predictor. This means that even if there is a correlation between a variable and f_{esc}, that variable may still be assigned a low importance. This may explain why R_{UV} is assigned a relatively weak importance in our models relative to observational works (Mascia et al. 2023; Jaskot et al. 2024). R_{UV} has been shown to heavily trace a galaxy's star-formation history in THESAN-ZOOM (McClymont et al. 2025a), and so most of its predictive power may be simply in tracing star formation which is better encapsulated by SFR_{10}/SFR_{100} and ΔMS_{10}. Our inclusion of R_{SFR}, which shows much greater importance than R_{UV}, suggests that it is the more fundamental morphological property related to f_{esc}. Therefore, much of the observed correlation between R_{UV} and f_{esc} may be due to both R_{UV} and f_{esc} being more fundamentally affected by a galaxy's star-formation history.

Although stellar metallicity Z is not a dominant diagnostic in the models, it remains positively correlated with f_{esc} and $\dot{N}_{ion,esc}$, consistent with Kostyuk et al. (2023), who attribute this to longer-lived, metal-rich stars whose LyC emission is more likely to persist after their birth clouds have dispersed. The gas-to-stellar mass ratio is a proxy for both LyC absorption (via gas content M_{gas}) and LyC production (through its M_* dependence), explaining why it is one of the strongest indicators in our best f_{esc} predictor Model A (see Fig. 3). A similar reasoning applies to M_*/M_{vir}, though its lack of direct sensitivity to LyC absorption makes it a weaker diagnostic. Likewise, we find $L_{UV}/L_{H\alpha}$ to be a relevant indicator of f_{esc} since Hα traces ionizing photon absorption, while L_{UV} is correlated with their production. We see that many of the features identified as important by the random forests are correlated, indicating some level of redundancy among the selected features. This may explain why models C and D exhibit only slightly worse performance than models A and B, respectively, across the metrics reported in Tab. 2, despite relying on more restrictive feature sets.

We find that redshift plays a relatively minor role in determining f_{esc} which is in agreement with the findings of Papovich et al. (2025); we also find an even weaker redshift dependence in $\dot{N}_{ion,esc}$. However, this likely reflects that any redshift evolution in properties, such as the merger rate or galaxy burstiness (McClymont et al. 2025b), is already captured by other predictors (e.g., SFR_{10}/SFR_{100}) that more directly measure these physical drivers of escape. The relative unimportance

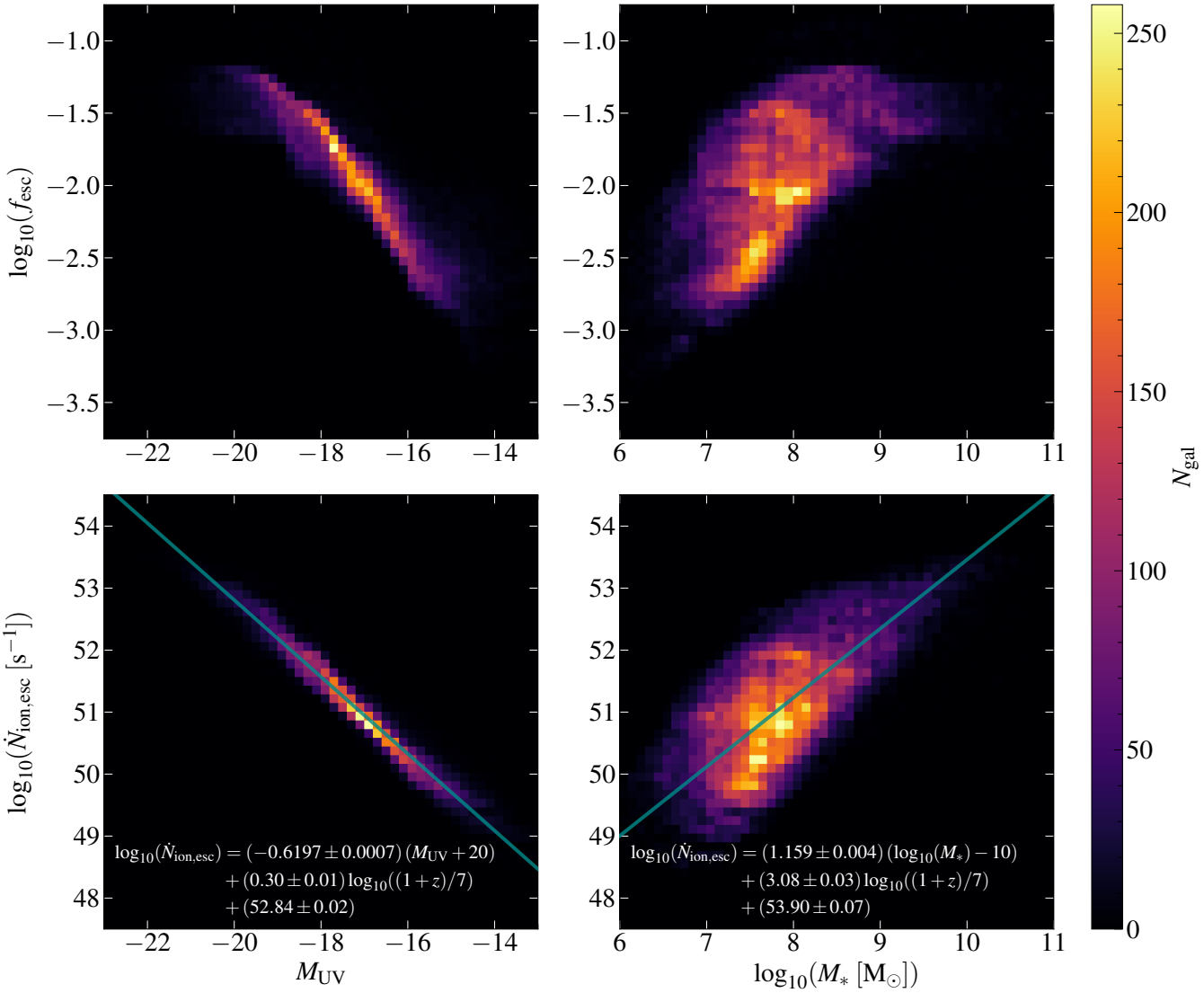


Figure 5. Model predictions of $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$ for galaxies in the JWST photometric catalogue from Simmonds et al. (2025). *Top left:* Predicted $f_{\rm esc}$ as a function of $M_{\rm UV}$. *Top right:* predicted $f_{\rm esc}$ as a function of M_*. *Bottom left:* predicted $\dot{N}_{\rm ion,esc}$ as a function of $M_{\rm UV}$. *Bottom right:* predicted $\dot{N}_{\rm ion,esc}$ as a function of M_*. The colour scale indicates the number of galaxies per bin. Both $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$ show clear correlations with $M_{\rm UV}$ and M_*, consistent with expectations from THESAN-ZOOM. In particular, $\log_{10}(\dot{N}_{\rm ion,esc})$ exhibits approximately linear behaviour with both $M_{\rm UV}$ and $\log_{10}(M_*)$. Therefore, in both lower panels, we present the orthogonal distance regression linear relations for $\log_{10}(\dot{N}_{\rm ion,esc})$, in which we also include a redshift dependence, and plot them in cyan. The relationship between predicted $\dot{N}_{\rm ion,esc}$ and $M_{\rm UV}$ exhibits very little scatter, implying that the model relies most strongly on $M_{\rm UV}$ in prediction, in agreement with the importance analysis presented in Fig. 3.

of redshift therefore suggests that there is little redshift evolution relevant for ionizing escape in variables not accounted for here.

3.4.2 $\dot{N}_{\rm ion,esc}$ *indicator physics*

A galaxy's $\dot{N}_{\rm ion,esc}$ is defined as its total UV luminosity multiplied by $f_{\rm esc}$ and its ionizing photon production efficiency $\xi_{\rm ion}$. Observations indicate $\xi_{\rm ion}$ varies relatively weakly across the high-redshift population (Simmonds et al. 2024a,b; Papovich et al. 2025; Llerena et al. 2025; Austin et al. 2025), with Llerena et al. (2025) finding a mean value of $\log_{10}(\xi_{\rm ion}\,[{\rm Hz\,erg^{-1}}]) = 25.22$ with an observed scatter of 0.42 dex in a JWST/NIRSpec spectroscopic sample of 761 galaxies at $4 \leq z \leq 10$. We therefore expect extensive observables highly correlated with luminosity to dominate predictions of $\dot{N}_{\rm ion,esc}$ (such as M_* and stellar formation rates). This explains why $M_{\rm UV}$, a logarithmic reparametrization of the UV luminosity, is the most important feature in both the relevant Models (B and D). It also explains why ratio-based features, which remove the leading-order multiplicative dependence on luminosity by probing relative rather than absolute quantities, as well as morphological parameters, contribute little additional predictive power once the dominant luminosity scaling is accounted for. The greater importance of $\rm SFR_{100}$ than $\rm SFR_{10}$ in predicting $\dot{N}_{\rm ion,esc}$ (see Fig. 3) supports the delayed-escape picture discussed in Section 3.4.1, with the longer-timescale formation rate observable providing greater predictive power.

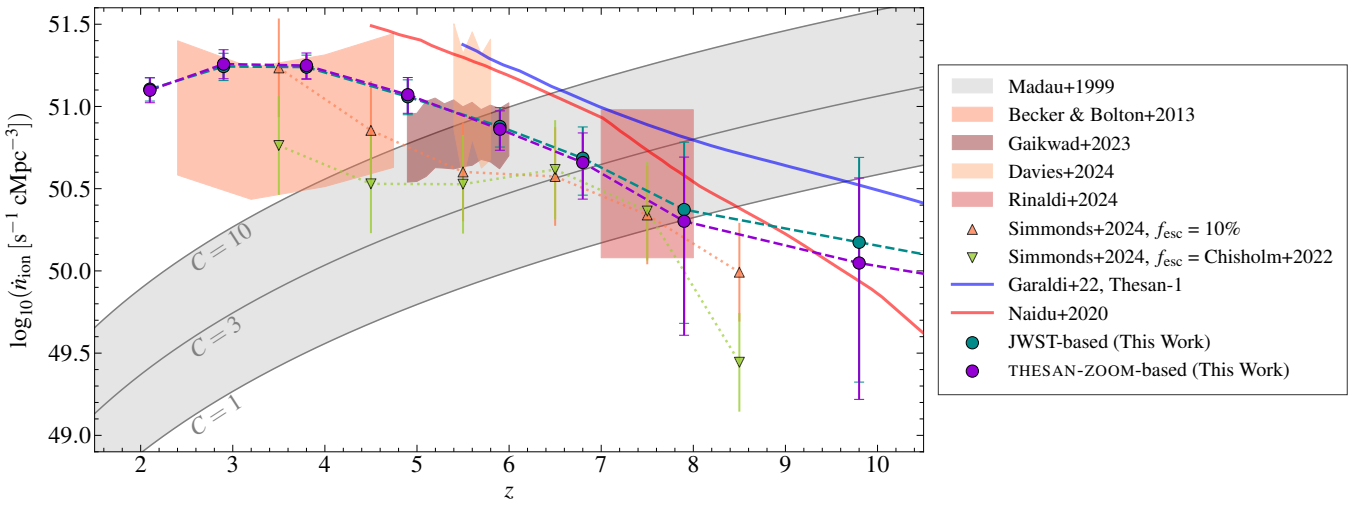


Figure 6. Comoving cosmic ionizing photon emissivity, $\dot{n}_{\rm ion}$, as a function of redshift, z, derived from the THESAN-ZOOM-based (violet points) and JWST-based (cyan points) $\dot{N}_{\rm ion,esc}$ relations. These emissivities have been integrated using the UV luminosity functions from Bouwens et al. (2021) for $z \in [1.5, 8.5]$ and from Whitler et al. (2025) for $z \in [8.5, 16]$. The grey bands indicate the critical ionizing emissivity required to maintain hydrogen ionization in the IGM, determined using the recombination model of Madau et al. (1999), for possible clumping factors $C \in [1, 10]$. For comparison, we include the $\dot{n}_{\rm ion}$ estimates of Simmonds et al. (2024b), derived from a related JWST/NIRCam-detected JADES sample assuming either a constant $f_{\rm esc} = 10\%$ or the $f_{\rm esc}$ model of Chisholm et al. (2022) (shown by the orange and green points respectively). We also include the curve derived from the parent THESAN-1 simulation run, as presented by Garaldi et al. (2022) (blue), and that of Naidu et al. (2020), derived from an empirical galaxy formation model with a fitted escape fraction (red). Observational constraints obtained from Lyα forest measurements and JWST observations of Hα emitters are overplotted (Becker & Bolton 2013; Gaikwad et al. 2023; Davies et al. 2024; Rinaldi et al. 2024). Both of our $\dot{n}_{\rm ion}$ histories are mutually consistent across all relevant redshifts and agree with the observational constraints. These results therefore avoid any crisis in the ionizing photon budget as they predict the critical emissivity is unlikely to have been reached until after $z \sim 8$.

4 IMPLICATIONS FOR COSMIC REIONIZATION

Recent JWST studies have reported a high abundance of UV bright early-Universe galaxies (Carniani et al. 2024; Donnan et al. 2024; Robertson et al. 2024; Tacchella 2024; Whitler et al. 2025), raising concerns about a potential "crisis in the ionizing photon budget" in which reionization appears to reach completion too early, in conflict with existing observations (Chakraborty & Choudhury 2024; Muñoz et al. 2024; Simmonds et al. 2024b; Bera et al. 2025). Here we discuss the implications for the evolution of reionization inferred from both the THESAN-ZOOM simulations and our predictions for the JWST/NIRCam-detected JADES sample of Simmonds et al. (2025), within the context of current observational constraints.

4.1 Ionizing emissivity

From Figs. 3 and 5, it is evident that $M_{\rm UV}$ is the most important quantity for constraining $\dot{N}_{\rm ion,esc}$. Fig. 5 also demonstrates that a linear relationship between $\log_{10}(\dot{N}_{\rm ion,esc})$ and $M_{\rm UV}$ is appropriate. This motivates us to use the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relations we have derived from both our JWST sample predictions and the THESAN-ZOOM simulations to obtain the corresponding histories of the cosmic ionizing emissivity as a function of redshift, $\dot{n}_{\rm ion}(z)$, through Eq. (1). This quantity, often referred to as the cosmic ionizing photon budget, allows us to investigate the implications of our relations for reionization. The resulting $\dot{n}_{\rm ion}(z)$ histories are shown in Fig. 6.

The $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation obtained from application of our observational $\dot{N}_{\rm ion,esc}$ predictor (model D) to the JWST sample is shown in the lower-left panel of Fig. 5 and is restated here:

$$\log_{10}(\dot{N}_{\rm ion,esc}) = (-0.6197 \pm 0.0007)\,(M_{\rm UV} + 20) + (0.30 \pm 0.01)\,\log_{10}\left(\frac{1+z}{7}\right) + (52.84 \pm 0.02)\,. \quad (6)$$

We also state the corresponding relation derived from the THESAN-ZOOM simulations (which is analogously presented in Fig. A4 in Appendix A):

$$\log_{10}(\dot{N}_{\rm ion,esc}) = (-0.670 \pm 0.002)\,(M_{\rm UV} + 20) + (0.28 \pm 0.03)\,\log_{10}\left(\frac{1+z}{7}\right) + (52.84 \pm 0.07)\,. \quad (7)$$

We note that we have included all galaxies in our catalogue when carrying out this fit, including those with $\mathrm{SFR}_{50} = 0$, which were neglected in the previous sections. This is because, despite typically having low luminosities, these galaxies may still be observable and included in the UVLF, thereby influencing the fitted relation. We additionally restrict the sample to galaxies with $M_{\rm UV} < -13$. Our THESAN-ZOOM catalogue includes only systems resolved with at least 100 stellar particles, imposing a stellar-mass completeness limit of $M_* \approx 9 \times 10^5\,\mathrm{M}_\odot$; with this mass scale roughly corresponding to a UV magnitude of $M_{\rm UV} \sim -13$. The sample is therefore incomplete at fainter magnitudes, motivating the aforementioned restriction.

We use the UVLFs from Bouwens et al. (2021) for $z \in [1.5, 8.5]$ and from Whitler et al. (2025) for $z \in [8.5, 16]$, which we parametrize in the Schechter form (Schechter 1976). This choice is motivated in Appendix C, where we compare the ionizing emissivities resulting from the Schechter and Double Power Law (DPL) parametrizations of the UVLF (Bowler et al. 2020; Whitler et al. 2025), as well as the

Schechter parametrization of the stellar mass function (Weibel et al. 2024), and demonstrate that the UVLF in Schechter form is most suitable for this work. Uncertainties in our $\dot{n}_{\rm ion}$ values are primarily driven by the reported uncertainties in these Schechter parameters.

In Fig. 1 we see that there is an approximately Gaussian scatter in $\log_{10}(\dot{N}_{\rm ion,esc})$ about the median $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation for the THESAN-ZOOM galaxies. Since the logarithm is nonlinear, $\langle \log_{10} \dot{N}_{\rm ion,esc} \rangle \neq \log_{10} \langle \dot{N}_{\rm ion,esc} \rangle$; consequently, Gaussian scatter in $\log_{10}(\dot{N}_{\rm ion,esc})$ boosts the linear-space mean escape rate, as upward fluctuations are weighted more strongly than downward fluctuations. As the relations stated in Eqs. (6) and (7) trace the mean of $\log_{10}(\dot{N}_{\rm ion,esc})$, we therefore multiply them by the standard lognormal correction factor to recover $\langle \dot{N}_{\rm ion,esc} \rangle$ for use in Eq. (1) (Shen et al. 2023; Jun et al. 2025):

$$\langle \dot{N}_{\rm ion,esc} \rangle = 10^{\frac{1}{2}\ln(10)\,\sigma^2_{\dot{N}}}\, \dot{N}_{\rm ion,esc}. \tag{8}$$

Here $\sigma_{\dot{N}}$ is taken to be half the $16^{\rm th} - 84^{\rm th}$ percentile range of the residuals in the THESAN-ZOOM $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation, which gives $\sigma_{\dot{N}} = 0.725$ dex. We note that this differs from the value of $\sigma_{\dot{N}}$ quoted in Section 3.1, due to the aforementioned exclusion in that section of THESAN-ZOOM galaxies with $\mathrm{SFR}_{50} = 0$. The validity of assuming a constant scatter in this relation is discussed in Appendix C. As shown in Fig. 5, our JWST sample predictions of $\dot{N}_{\rm ion,esc}$ are very strongly constrained by $M_{\rm UV}$, artificially suppressing the intrinsic scatter. We therefore adopt the THESAN-ZOOM value of $\sigma_{\dot{N}}$ for both the THESAN-ZOOM- and JWST-based calculations of $\dot{n}_{\rm ion}$. We discuss the consequences of different $\sigma_{\dot{N}}$ on reionization histories in Section 4.5.

Eq. (1) converges at the high luminosity limit and is therefore insensitive to the choice of $M_{\rm UV,min}$. However, the integral diverges slowly toward faint galaxies, requiring a finite $M_{\rm UV,max}$ cutoff when evaluating $\dot{n}_{\rm ion}$. Consistent with the completeness limit of our THESAN-ZOOM catalogue discussed above, we adopt $M_{\rm UV,max} = -13$, which is in agreement with values assumed in previous studies (e.g. Robertson et al. 2015; Larson et al. 2022). In Section 4.5, we examine the sensitivity of our results to the choice of $M_{\rm UV,max}$ and show that our inferred reionization histories depend only weakly on this assumption.

Strong constraints on $\dot{n}_{\rm ion}$ have been placed by the Lyα forest at lower redshifts ($z < 6$); accordingly, we include the observational results of Becker & Bolton (2013), Gaikwad et al. (2023) and Davies et al. (2024) in Fig. 6. From such measurements, $\dot{n}_{\rm ion}$ has been observed to flatten at the end of the EoR with a value of $\log_{10}(\dot{n}_{\rm ion}\ [\mathrm{s^{-1}\ cMpc^{-3}}]) \approx 51$ at $z \lesssim 5-6$ (Becker & Bolton 2013; Davies et al. 2024). Our THESAN-ZOOM-based and JWST-based $\dot{n}_{\rm ion}$ histories agree across all redshifts. Furthermore, both our $\dot{n}_{\rm ion}(z)$ histories are consistent with these constraints and with the expected post-reionization flattening. Although early-Universe $\dot{n}_{\rm ion}$ values are highly uncertain, we also include the estimates of Rinaldi et al. (2024), inferred from JWST measurements of strong Hα emitters for $z \in [7, 8]$, and find that our results are in good agreement with these as well.

For comparison to the two $\dot{n}_{\rm ion}$ curves of this work in Fig. 6, we add two more from Simmonds et al. (2024b), derived via PROSPECTOR-inferred $\xi_{\rm ion} - M_{\rm UV}$ relations of a related JWST sample based on NIRCam imaging of the JADES in the GOODS-South region and the UVLFs of Bouwens et al. (2021). The first assumes a constant LyC escape fraction of $f_{\rm esc} = 10\%$, while the second employs the UV continuum slope-dependent empirical $f_{\rm esc}$ derived by (Chisholm et al. 2022) from a low-redshift sample. The $\dot{n}_{\rm ion}$ histories of this work show broad consistency with the results of Simmonds et al. (2024b) across redshifts relevant to cosmic reionization ($z \gtrsim 5$), with closer agreement to the constant $f_{\rm esc}$ curve than to the curve using an $f_{\rm esc}$ prescription derived from a low-redshift analogues. We also include the $\dot{n}_{\rm ion}$ curve derived by Garaldi et al. (2022) from the parent THESAN simulations for the THESAN-1 run. While somewhat consistent with our results at high redshifts, the $\dot{n}_{\rm ion}$ inferred from THESAN diverges from Lyα forest observational constraints at $z \lesssim 6$, particularly the Lyα opacity constraints of Gaikwad et al. (2023). The improved resolution and physical modeling of the THESAN-ZOOM simulations could plausibly explain why our results remain consistent with observational constraints, highlighting the advantage of the zoom-in technique for determining the ionizing emissivity. A similar behaviour is seen in the $\dot{n}_{\rm ion}$ curve inferred by Naidu et al. (2020), based on the empirical galaxy formation model of Tacchella et al. (2018) with SED-computed ionizing photon production efficiencies and a fitted escape fraction, which likewise departs from the observational constraints of Gaikwad et al. (2023). Austin et al. (2025) measure a constant $\xi_{\rm ion}$ with UV magnitude for a JWST/NIRCam sample across $5.6 < z < 6.5$ galaxies, and construct Schechter UVLFs from the same sample. For $M_{\rm UV,max} = -13.5$, and adopting the same $f_{\rm esc}$ prescriptions as Simmonds et al. (2024b), Austin et al. (2025) obtain $\log_{10}(\dot{n}_{\rm ion}\ [\mathrm{s^{-1}\ cMpc^{-3}}]) = 51.0 \pm 0.2$ assuming $f_{\rm esc} = 10\%$ and $\log_{10}(\dot{n}_{\rm ion}\ [\mathrm{s^{-1}\ cMpc^{-3}}]) = 50.9 \pm 0.2$ assuming the $f_{\rm esc}$ calibration of Chisholm et al. (2022). Their results are consistent with our $\dot{n}_{\rm ion}$ values at $z \approx 6$.

We include curves in Fig. 6 quantifying the critical ionizing emissivity, $\dot{n}^*_{\rm ion}(z)$, required to balance the IGM recombination rate and maintain the full ionization of hydrogen at z. It is given as the ratio of the mean comoving cosmic number density of hydrogen atoms, $\langle n_{\rm H} \rangle = 1.88 \cdot 10^{-7}\,\mathrm{cm^{-3}}$ (Ade et al. 2016), to the effective IGM HII recombination timescale at z, $\bar{t}_{\rm rec}(z)$ (Madau et al. 1999). We compute this effective timescale $\bar{t}_{\rm rec}$ using the model of Madau et al. (2024):

$$\frac{1}{\bar{t}_{\rm rec}(z)} = (1+\chi)\,\alpha_B(T_0)\,\langle n_{\rm H} \rangle\,(1+z)^3\,C\,, \tag{9}$$

where $\alpha_B = 2.58 \cdot 10^{-13}\,\mathrm{cm^3\,s^{-1}}$ is the recombination coefficient at a fixed temperature of $T_0 = 10^4\,\mathrm{K}$ (Madau et al. 2024), and $\chi \equiv Y/4X = 0.0819$ which assumes helium is only singly ionized at the same time as hydrogen (Ade et al. 2016). Finally, C is the clumping factor of ionized hydrogen and accounts for the density inhomogeneities in the IGM; it is poorly constrained (Frenk et al. 2000; So et al. 2014), so we plot curves for clumping factors of 1, 3 and 10 in Fig. 6. Our $\dot{n}_{\rm ion}$ results predict that critical emissivity is unlikely to have been reached until after $z \sim 8$, with reionization completing at a much lower redshift; this is supported by our analysis of the evolution of the hydrogen ionization fraction in Section 4.3. Consequently, this work avoids the overabundance of ionizing photons predicted in recent JWST studies (Chakraborty & Choudhury 2024; Muñoz et al. 2024; Austin et al. 2025), without the need to assume a high clumping factor. We note that Papovich et al. (2025) find that a lower average escape fraction than that generally predicted at low redshifts can also reconcile the ionizing photon budget with constraints.

4.2 Galaxy Population Contributions to Reionization

We can split our $\dot{n}_{\rm ion}(z)$ histories into contributions from different UV magnitude populations to determine which sources drove reionization. For both the THESAN-ZOOM-based and JWST-based analyses, the integration of Eq. (1) is performed separately over three $M_{\rm UV}$ bins: $-13 \geq M_{\rm UV} > -17$, $-17 \geq M_{\rm UV} > -20$, and $M_{\rm UV} \leq -20$.

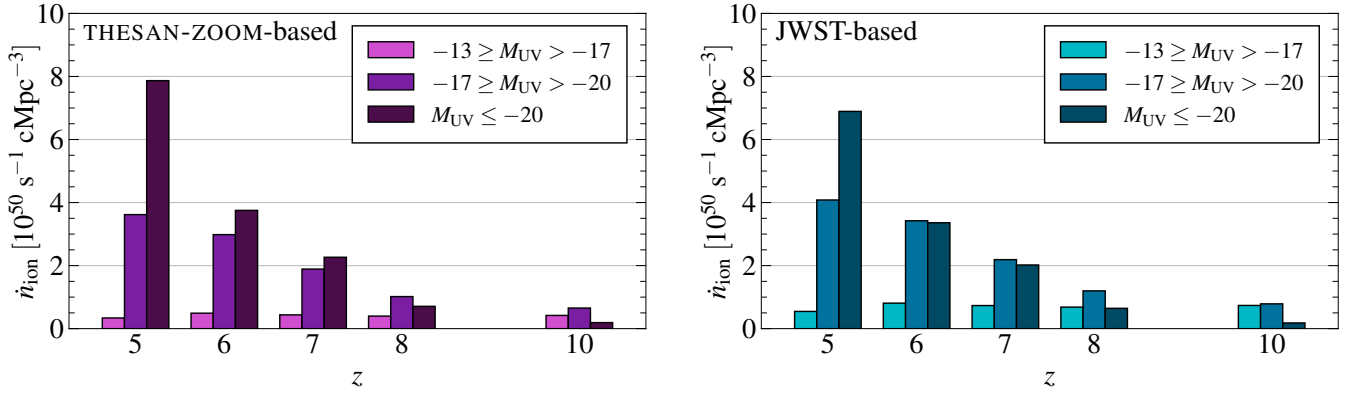


Figure 7. Contributions to the cosmic ionizing emissivity, $\dot{n}_{\rm ion}$, from galaxies in three separate UV magnitude bins: $-13 \geq M_{\rm UV} > -17$, $-17 \geq M_{\rm UV} > -20$ and $M_{\rm UV} \leq -20$. *Left panel:* $\dot{n}_{\rm ion}$ derived using the THESAN-ZOOM-based $\dot{N}_{\rm ion,esc}$ relation (Eq. (7)), as shown in Fig. 6, decomposed into contributions from the UV magnitude bins. *right panel:* $\dot{n}_{\rm ion}$ derived using the JWST-based $\dot{N}_{\rm ion,esc}$ relation (Eq. (6)), as shown in Fig. 6, decomposed into contributions from the UV magnitude bins. For $z < 8.5$ we adopt the UVLF of Bouwens et al. (2021), which is reported in redshift bins of width $\Delta_z = 1$. For $z > 8.5$ we use a UVLF from Whitler et al. (2025), measured over the broad interval $z \in [8.5, 12]$ and reported at its bin-averaged redshift $\langle z \rangle \approx 10$, hence no emissivity values are shown at $z = 9$ here. At high redshifts, both analyses show an approximately egalitarian picture of reionization, with neither the faintest galaxies nor the brightest galaxies dominating the $\dot{n}_{\rm ion}$ budget. However, both histories find that brighter galaxies contribute increasingly more toward lower redshifts, with the $M_{\rm UV} \leq -20$ population providing the greatest contribution to the ionizing photon budget by the completion of reionization.

The resulting contributions to the ionizing photon budget are presented in the left and right panels of Fig. 7, respectively. This figure paints an egalitarian picture of reionization at high redshifts, with neither the faintest nor the brightest galaxies dominating the $\dot{n}_{\rm ion}$ budget. However, for $z \lesssim 8$, both analyses convey that brighter galaxies contribute increasingly more, with these luminous giants driving reionization by its completion. The THESAN-ZOOM-based analysis, for example, predicts that the $M_{\rm UV} \leq -20$ population provides more than 50% of the ionizing photon budget by $z \sim 6$.

These findings are in agreement with the conclusions of Naidu et al. (2020), whose $\dot{n}_{\rm ion}$ curve is shown in Fig 6, and others (Larson et al. 2022; Roberts-Borsani et al. 2023; Yeh et al. 2023; Bera et al. 2025), but challenges alternative recent studies that attribute reionization primarily to the hidden faint-end population (Dayal et al. 2020; Atek et al. 2024; Mascia et al. 2024; Simmonds et al. 2024b). Consistent with our findings, Bera et al. (2025) argue that a substantial contribution from very bright galaxies, increasing toward lower redshifts, is required to reproduce JWST observations while avoiding a history of reionization that ends too early. Similarly, Jecmen et al. (2026) report a subdominant contribution from dim galaxies with $M_{\rm UV} > -14$, driven by declining escape fractions toward the faint end, in agreement with our findings.

Simmonds et al. (2024b) use SED fitting with PROSPECTOR to derive $\dot{n}_{\rm ion} - M_{\rm UV}$ relations for their related JWST sample; finding slopes in the range -0.47 to -0.39, which are noticeably more gentle than the values of -0.6197 and -0.670 inferred in this work (see Eq. (6) and (7)). The steeper relations obtained here imply a larger ionizing-photon output from brighter galaxies, resulting in a significantly greater role for bright galaxies in reionization.

4.3 Hydrogen ionization fraction

As in Wu et al. (2025), we use our $\dot{n}_{\rm ion}(z)$ curves to solve for the volume-averaged hydrogen ionized fraction $Q_{\rm HII}$ (Madau et al. 1999), employing the identity $dt = -dz\,(H(z)\,(1+z))^{-1}$ to re-write $Q(t)$ as a function of redshift (Wu et al. 2025):

$$\begin{aligned}\frac{dQ_{\rm HII}}{dz} &= -\frac{1}{H(z)\,(1+z)}\,\frac{dQ_{\rm HII}}{dt} \\ &= -\frac{1}{H(z)\,(1+z)}\left(\frac{\dot{n}_{\rm ion}}{\langle n_{\rm H}\rangle} - \frac{Q_{\rm HII}}{\bar{t}_{\rm rec}}\right), \end{aligned} \tag{10}$$

where we assume a flat cosmology with the cosmological parameters of the Planck Collaboration (Ade et al. 2016). The relevant recombination timescale is given in Eq. (9) and is dependent on the ionized hydrogen clumping factor C. In this analysis, we follow the IGM clumping model used by Madau et al. (2024) and Wu et al. (2025), which employs the parametrization of $C = 9.25 - 7.21\log_{10}(1+z)$ for the clumping factor, calibrated from the radiation-hydrodynamic simulations of ionization inhomogeneities of Finlator et al. (2009) and Chen et al. (2020).

We interpolate both our THESAN-ZOOM-based and JWST-based $\dot{n}_{\rm ion}(z)$ histories using weighted cubic splines and integrate Eq. (10) to a given z from a fully neutral Universe $Q_{\rm HII}(z_i) = 0$ at $z_i = 20$ (Virtanen et al. 2020), with the resulting ionization fraction evolutions shown in Fig. 8. We include in this figure observational constraints placed on $Q_{\rm HII}$ from the CMB (Aghanim et al. 2020), quasar spectrum damping-wing analyses (Davies et al. 2018; Wang et al. 2020), Lyα emissions (Tang et al. 2024; Kageura et al. 2025) and the combined Lyα and Lyβ forests (Jin et al. 2023; Zhu et al. 2024). Our THESAN-ZOOM-based and JWST-based reionization histories are in good agreement with each other and these measurements. We find reionization reaches completion ($Q_{\rm HII}(z_f) = 1$) at $z_f = 5.5 \pm 0.2$ according to the THESAN-ZOOM-based results, while the JWST-based yields a comparable value of $z_f = 5.6 \pm 0.3$. Both are consistent with current observational evidence, which indicates completion in the range $5 < z_f < 6$ (Becker et al. 2021; Jin et al. 2023; Davies et al. 2024; Zhu et al. 2024). We note that AGN are neglected in the THESAN-ZOOM simulations, despite potentially contributing a non-negligible fraction of the ionizing photon budget, as discussed in Section 1. Importantly, we have again shown that the reionization results derived from our $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relations do not suffer from

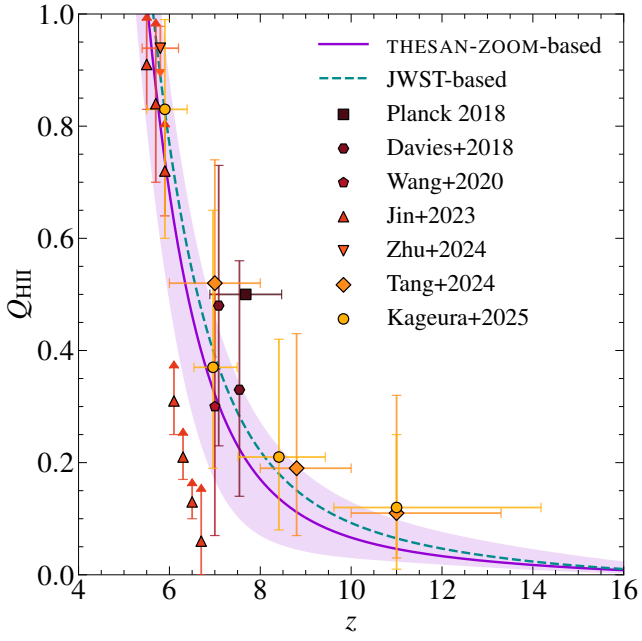


Figure 8. Hydrogen ionized fraction, $Q_{\rm HII}$, as a function of redshift, z, computed using Eq. (10) together with the THESAN-ZOOM-based (violet curve) and JWST-based (cyan curve) ionizing emissivities from Fig. 6. The shaded violet region indicates the uncertainty in $Q_{\rm HII}$ for the THESAN-ZOOM-based analysis (we have neglected the similar JWST-based uncertainty for figure clarity). Observational measurements and limits, represented by points and arrows, are plotted, including constraints from: the CMB (Aghanim et al. 2020), quasar spectrum damping-wing analyses (Davies et al. 2018; Wang et al. 2020), Lyα emissions (Tang et al. 2024; Kageura et al. 2025) and the combined Lyα and Lyβ forests (Jin et al. 2023; Zhu et al. 2024). Both the THESAN-ZOOM-based and JWST-based ionization fraction histories are consistent with all these constraints and each other. Furthermore, they both predict a rapid, late reionization process, with $Q_{\rm HII}$ likely less than 0.2 at $z = 8$.

the ionizing photon overabundance reported in other JWST studies (Muñoz et al. 2024).

In both our ionization fraction evolutions, we find $Q_{\rm HII}(z = 8) \leq 0.2$, implying the bulk of reionization occurred rapidly after $z \sim 8$. This is in agreement with other recent works (Kulkarni et al. 2019; Naidu et al. 2020; Choustikov et al. 2024c; Jecmen et al. 2026), with Kulkarni et al. (2019) arguing that a rapid evolution is required to reproduce Lyα constraints. Naidu et al. (2020), who, as previously discussed, also find that the most luminous galaxies account for the majority of the ionizing photon budget, conclude that the increased population of such massive galaxies at lower redshifts causes this accelerated end to reionization.

4.4 Thomson optical depth

The Thomson optical depth, τ, quantifies the scattering of CMB photons by free electrons produced during the EoR; it therefore provides a useful constraint on reionization. Using the hydrogen ionized fractions $Q_{\rm HII}(z)$ shown in Fig. 8, $\tau(z)$ can be computed as (Wu et al. 2025):

$$\tau(z) = c\,\sigma_T\,\langle n_{\rm H}\rangle \int_0^z (1+\eta\,\chi)\,\frac{(1+z')^2}{H(z')}\,Q_{\rm HII}(z')\,{\rm d}z'\,, \qquad (11)$$

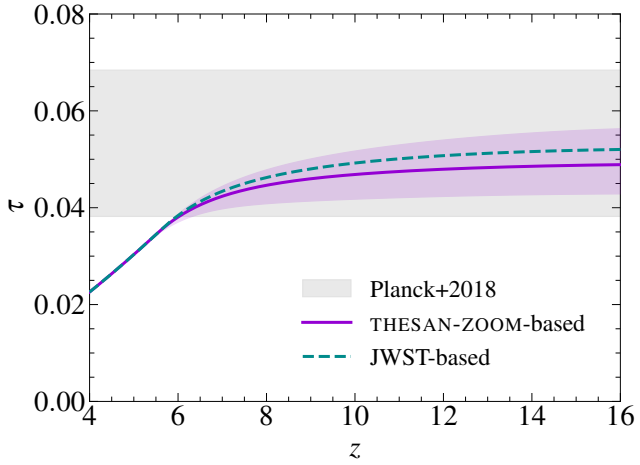


Figure 9. Thomson optical depth, τ, as a function of redshift, z, computed using Eq. (11) together with the THESAN-ZOOM-based (violet curve) and JWST-based (cyan based) $Q_{\rm HII}$ histories from Fig. 8. The shaded violet region indicates the uncertainty in τ for the THESAN-ZOOM-based analysis (we have neglected the similar JWST-based uncertainty for figure clarity). The grey band shows the 2σ range of the opacity constraints measured by The 2018 Planck collaboration (Aghanim et al. 2020). Both the THESAN-ZOOM-based and JWST-based $\tau(z)$ asymptote well within this range. Thus, the optical depths inferred from our $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relations are consistent with constraints from the CMB.

where c is the speed of light, σ_T is the Thomson scattering cross-section, and the prefactor η accounts for the free electron contribution from helium, assuming helium is singly ionized for $z > 4$ ($\eta = 1$) and fully ionized at lower redshifts ($\eta = 2$).

Fig. 9 shows $\tau(z)$ derived from both our THESAN-ZOOM-based and JWST-based $Q_{\rm HII}(z)$ curves. We compare the high-redshift asymptotes of our predicted optical depths with the measurement $\tau_{\rm CMB} = 0.054^{+0.014}_{-0.016}$ (2σ uncertainties) from the Planck Collaboration (Aghanim et al. 2020). Some recent studies have noted a potential tension between the baryon acoustic oscillation and the Planck CMB measurements within the ΛCDM framework that could be alleviated if $\tau_{\rm CMB}$ were underpredicted (Jhaveri et al. 2025; Liu et al. 2025; Sailer et al. 2025). However, it has been shown that the previously discussed constraints on the hydrogen neutral fraction from the Lyα forest and damping-wing absorption, together with limits on the patchy kinematic Sunyaev–Zel'dovich signal, which is a CMB probe of inhomogeneous reionization (Chen et al. 2023), require reionization histories that yield optical depths consistent with the Planck value (Cain et al. 2025; Kageura et al. 2026).

Both optical depths derived in this work (from the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relations found in THESAN-ZOOM and our JWST sample $\dot{N}_{\rm ion,esc}$ predictions) are consistent with this CMB constraint, with uncertainty ranges that closely overlap those of the Planck result. This agreement further supports both the validity of the THESAN-ZOOM simulations and the predictive models we presented in Section 3.

4.5 Uncertainties in the Reionization Histories

In computing the ionizing emissivities of Fig. 6, we adopt $M_{\rm UV,max} = -13$ as the faint-end cutoff of Eq. (1). This choice was well motivated as the THESAN-ZOOM catalogue is approximately complete down to this luminosity. However, the selection of faint-end cutoffs varies across the literature (Naidu et al. 2020; Larson et al. 2022; Sim-

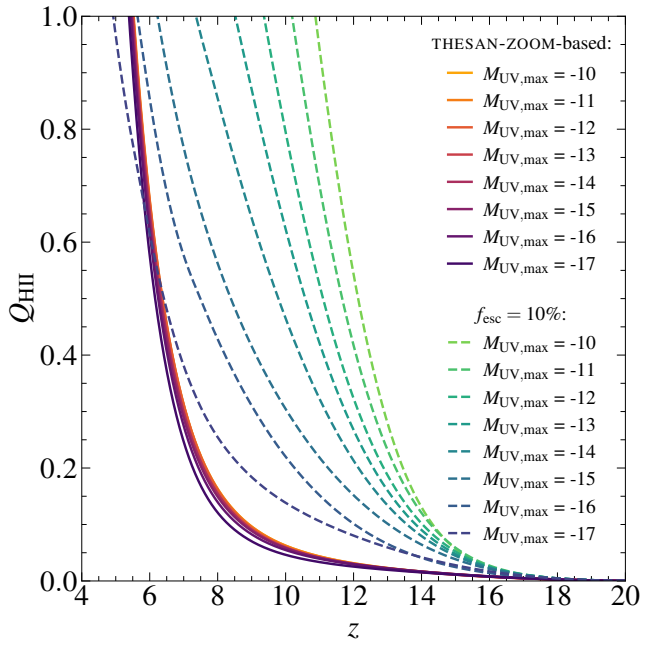


Figure 10. Hydrogen ionized fraction, $Q_{\rm HII}$, as a function of redshift, z, calculated from ionizing emissivities using different choices of the faint-end cutoff in the integral of Eq. (1), with $M_{\rm UV,max}$ spanning $[-10, -17]$. Solid lines are derived from the THESAN-ZOOM-based $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation given in Eq. 7, while dashed lines are obtained assuming a constant ionizing photon production efficiency of $\log_{10}(\xi_{\rm ion}\,[{\rm Hz\,erg^{-1}}]) = 25.05$ (Austin et al. 2025), and a constant ionizing photon escape fraction of $f_{\rm esc} = 10\%$. We find that the evolution of reionization in this work is largely insensitive to the choice of the faint-end cutoff. However, this choice has a substantial impact on both the evolution and completion of reionization in models that assume a constant ionizing photon production efficiency and a constant ionizing photon escape fraction.

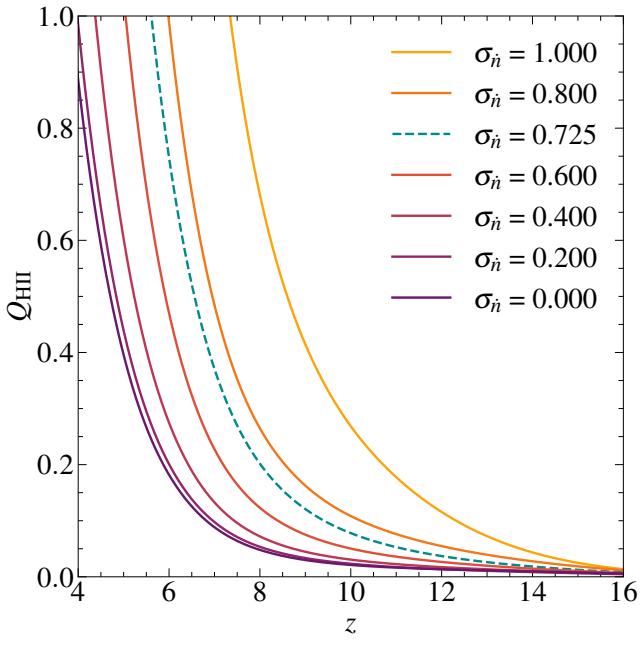


Figure 11. Hydrogen ionized fraction, $Q_{\rm HII}$, as a function of redshift, z, for JWST-based $\dot{n}_{\rm ion}$ histories. The unbroken curves show results obtained by assuming $\sigma_{\dot{N}} \in [0.000, 1.000]$ dex, corresponding to a range of possible scatter in the JWST-based $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation of Eq. (6), when applying the lognormal correction factor described in Section 4.1. The fiducial curve, with $\sigma_{\dot{N}} = 0.725$ corresponding to the scatter in the THESAN-ZOOM relation, is also included in cyan. Even modest increases in stochasticity produce significant shifts in the redshift at which reionization is predicted to complete.

monds et al. 2024b; Papovich et al. 2025; Wu et al. 2025; Austin et al. 2025). To investigate the impact of the choice of $M_{\rm UV,max}$ on our reionization histories, we repeat the THESAN-ZOOM-based computation of the hydrogen ionized fraction, $Q_{\rm HII}$, shown in Fig. 8, using a range of $\dot{n}_{\rm ion}$ histories calculated with faint-end cutoffs $M_{\rm UV,max}$ spanning $[-10, -17]$, with the resulting ionization histories shown in Fig 10. We find that the evolution of reionization is largely insensitive to the choice of this integration limit, with changes in $M_{\rm UV,max}$ of ± 3 magnitudes about the adopted value of -13 resulting in a shift in the redshift of reionization completion of less than 0.1. This behaviour likely reflects the relatively minor contribution of the more numerous populations of faint galaxies to the overall ionizing photon budget inferred in this work. For comparison, we plot the evolution of $Q_{\rm HII}$ across the same range of $M_{\rm UV,max}$, for a model that assumes a constant ionizing photon production efficiency of $\log_{10}(\xi_{\rm ion}\,[{\rm Hz\,erg^{-1}}]) = 25.05$) (Austin et al. 2025), and a constant ionizing photon escape fraction of $f_{\rm esc} = 10\%$, such that $\dot{N}_{\rm ion,esc}$ depends solely on galactic UV luminosity. We see that the evolution of reionization for such a model is highly dependent on the choice of faint-end cutoff. Such assumptions of a constant escape fraction are not uncommon in recent studies (Simmonds et al. 2024a,b; Wu et al. 2025; Austin et al. 2025); consequently, ionizing emissivities inferred under assumptions of a constant escape fraction and a relatively bright faint-end cutoff should be interpreted as lower limits, as the contribution from the faintest galaxies is neglected. By contrast, the weak sensitivity to $M_{\rm UV,max}$ in our framework gives us confidence that our inferred $\dot{n}_{\rm ion}$ values are not systematically underpredicted.

In Section 4.1 we discussed how scatter in the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation introduces a systematic offset between the derived mean $\log_{10}(\dot{N}_{\rm ion,esc})$ relation and the mean ionizing photon escape rate, $\langle \dot{N}_{\rm ion,esc} \rangle$, requiring a correction factor when inferring $\dot{n}_{\rm ion}$. We do not have direct access to the intrinsic scatter of the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation of the JWST sample, since the $\dot{N}_{\rm ion,esc}$ values have been predicted using our feature-compatible $\dot{N}_{\rm ion,esc}$ model (Model D) presented in Section 3. Accordingly, our JWST-based reionization analysis assumed a scatter of $\sigma_{\dot{N}} = 0.725$ dex, equal to that of the THESAN-ZOOM relation. We investigate the significance of $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ intrinsic scatter to the volume-averaged hydrogen ionization fraction $Q_{\rm HII}$ in Fig. 11. The $Q_{\rm HII}(z)$ curves shown in this figure are derived using the same JWST-based $\dot{n}_{\rm ion}(z)$ framework as in Fig. 8, but with the assumed scatter of the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ JWST sample relation varied from 0.000 to 1.000 dex in steps of 0.200 dex. Fig. 11 reveals that even modest increases in relation variability can noticeably change the evolution of the EoR and significantly shift the redshift at which reionization completes, with z_f increasing from 3.8 to 7.3 over the $\sigma_{\dot{N}}$ range. In this work, we find the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation to be steep, and that reionization is driven primarily by UV-bright galaxies. In this regime, upward fluctuations toward higher $log_{10}(\dot{N}_{\rm ion,esc})$ values can substantially increase a galaxy's ionizing photon emission, while the steep decline in galaxy number density toward the bright end of the UVLF provides many more intermediate-luminosity galaxies that can scatter into high $\dot{N}_{\rm ion,esc}$ values than vice versa (Ren et al. 2019). Consequently, increasing the intrinsic scatter

strongly boosts the abundance of strong LyC emitters, leading to the large changes in $\dot{n}_{\rm ion}$ seen in Fig. 11. This reinforces the importance of accounting for intrinsic scatter when calculating $\dot{n}_{\rm ion}$ in reionization studies.

For the hydrogen recombination timescale in our $Q_{\rm HII}$ analysis, we have followed Wu et al. (2025) in adopting the redshift-dependent clumping factor C parametrization calibrated by Finlator et al. (2009) and Chen et al. (2020). However, the clumping factor is known to be poorly constrained (Frenk et al. 2000; So et al. 2014) and this parametrization should be interpreted with appropriate caution. To investigate how C affects reionization, we again integrate Eq. (10) using the THESAN-ZOOM-based $\dot{n}_{\rm ion}(z)$ curve of Fig. 6 (as done in Fig. 8), but now assume a constant IGM clumping factor of C = 1, 3, or 10. This yields reionization completion at $z_f = 5.8 \pm 0.3$, 5.5 ± 0.2, and 5.0 ± 0.2, respectively. As expected, a higher clumping factor delays reionization, as a larger number of ionizing photons is required to reionize the IGM (Wu et al. 2025). A perfectly uniform and homogeneous IGM ($C = 1$) is unrealistic, and Wu et al. (2025) suggests that an upper bound of $C \leq 10$ for all redshifts is likely an overestimate. Consequently, while the clumping factor is clearly relevant, its effect on reionization is less significant than that associated with variability in the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation.

Most recent studies of the EoR rely on a combination of models for both $f_{\rm esc}$ and either $\dot{N}_{\rm ion}$ or $\xi_{\rm ion}$ for predictions of the ionizing emissivity, $\dot{n}_{\rm ion}$, the hydrogen ionization fraction, $Q_{\rm HII}$, and the Thomson optical depth, τ. Many find that their predictions are highly sensitive to assumptions regarding the dependence of $f_{\rm esc}$ and $\xi_{\rm ion}$ on galaxy properties (Price et al. 2016; Naidu et al. 2020; Chisholm et al. 2022; Kostyuk et al. 2023; Simmonds et al. 2024a,b; Chakraborty & Choudhury 2024; Wu et al. 2025; Austin et al. 2025). Moreover, the challenges associated with reliably estimating $f_{\rm esc}$ have been discussed throughout this paper, and the predictive performance of our $f_{\rm esc}$ models is lower than that of our $\dot{N}_{\rm ion,esc}$ models. In contrast to previous works, we derive our reionization histories directly from linear fits to $\log_{10}(\dot{N}_{\rm ion,esc})$, thereby avoiding the need to separately model the poorly constrained quantities $f_{\rm esc}$ and $\xi_{\rm ion}$. This results in a simpler framework that nonetheless remains consistent with all the observational constraints on $\dot{n}_{\rm ion}$, $Q_{\rm HII}$, and τ shown in Figs. 6, 8, and 9.

5 CONCLUSIONS

We have combined high-resolution cosmological radiation-hydrodynamic simulations with the random forest machine learning technique to investigate the history of cosmic reionization predicted by state-of-the-art theoretical models in the context of the latest observational constraints. We have trained random forest regressor models on COLT-forward-modelled observables derived from over 35 000 galaxy realisations spanning $3 < z < 16$ in the THESAN-ZOOM simulations, to predict both $f_{\rm esc}$ (Model A) and $\dot{N}_{\rm ion,esc}$ (Model B). Our main findings are as follows:

- **Ionizing escape governed by bursty star formation.** We find the 10–to–100 Myr star-formation rate ratio ($\rm SFR_{10}/SFR_{100}$) and the gas-to-stellar mass ratio ($M_{\rm gas}/M_*$) to be the most important predictors of $f_{\rm esc}$. We interpret the leading diagnostic as a signature of bursty star-formation cycles in which feedback-driven gas outflows clear the ISM and facilitate LyC escape. Interestingly, $M_{\rm UV}$ is the third most important variable, likely because it traces total stellar mass, recent star-formation rate, and dust content, all of which are relevant to ionizing photon escape.
- **Distinct predictors for $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$.** In contrast to $f_{\rm esc}$, we find that $M_{\rm UV}$ dominates $\dot{N}_{\rm ion,esc}$ predictions, with star-formation rates (both $\rm SFR_{10}$ and $\rm SFR_{100}$) also relevant. This reflects the important role of the scaling of $\dot{N}_{\rm ion,esc}$ with stellar photon production as a galaxy grows and contains more stars. $M_{\rm UV}$ is therefore particularly powerful as it encapsulates not just information about this scaling, but also about the ionizing escape as discussed above.
- **Probing the sources of reionization seen by JWST.** We have created two further random forest models to predict $f_{\rm esc}$ (Model C) and $\dot{N}_{\rm ion,esc}$ (Model D) with exclusively photometrically-accessible galaxy properties. We have applied these models to a catalogue of JWST/NIRCam-detected galaxies to predict their ionizing properties, enabling us to derive the JWST-based $M_{\rm UV}$–$\dot{N}_{\rm ion,esc}$ relation for this observed sample. We find a steeper slope than in previous works, implying a larger contribution from bright galaxies to the cosmic ionizing photon budget.
- **Reionization models consistent with observations.** We use both the JWST-based $M_{\rm UV}$–$\dot{N}_{\rm ion,esc}$ relation and the relation directly measured from the simulations (THESAN-ZOOM-based) combined with observed UVLFs to obtain the cosmic ionizing emissivity, $\dot{n}_{\rm ion}$. When used to construct reionization histories, we find that both the JWST-based and THESAN-ZOOM-based models are consistent with observational constraints, with reionization completing at $z = 5.6 \pm 0.3$ and 5.5 ± 0.2, respectively, and avoid the "crisis in the ionizing photon budget" that has plagued some recent JWST studies.
- **Late reionization driven by bright sources.** Our results indicate that the bulk of reionization occurs rapidly after $z \sim 8$, driven by the most luminous galaxies, which dominate the cosmic reionization budget for $z \leq 8$.
- **Robustness to the poorly constrained UVLF faint-end.** Due to the dominance of bright galaxies, our model is remarkably robust to variations in the UVLF integration limits, in contrast to models of reionization dominated by faint dwarfs which are sensitive to the shape of the poorly-constrained faint end of the UVLF. However, we find that the intrinsic scatter of the $M_{\rm UV}$–$\dot{N}_{\rm ion,esc}$ relation has a strong impact on the timing of reionization.

Our work highlights the use of machine learning techniques to apply physical insights from smaller, high-resolution simulations to large statistical samples of galaxies. Furthermore, we have demonstrated the utility of directly modelling $\dot{N}_{\rm ion,esc}$ for statistical samples of galaxies, thereby avoiding the need to separately model the poorly constrained quantities $f_{\rm esc}$ and $\xi_{\rm ion}$.

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DATA AVAILABILITY

All simulation data, including snapshots, group, and subhalo catalogs and merger trees will be made publicly available in the near future. Data will be distributed via www.thesan-project.com. Before the public data release, data underlying this article will be shared on reasonable request to the corresponding authors. The public JADES data used are available from the Mikulski Archive for Space Telescopes (MAST; https://archive.stsci.edu/hlsp/jades), with doi:10.17909/8tdj-8n28, doi:10.17909/z2gw-mk31, and doi:10.17909/fsc4-dt61 (Rieke et al. 2023; Eisenstein et al. 2023; Bunker et al. 2024; D'Eugenio et al. 2025)

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APPENDIX A: THESAN-ZOOM POTENTIAL INDICATORS

We present the complete graphical analysis described in Section 3 to identify potential diagnostics of $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$ for all the variables in Tab. 1, justifying their inclusion in the list of indicators. Following Fig. 1, the single-variable correlation plots that establish a relationship between each variable and $f_{\rm esc}$ or $\dot{N}_{\rm ion,esc}$ are shown in Fig. A1 and A2 respectively. The ratio analysis, exemplified earlier in Fig. 2, is presented in full in Fig. A3. Correlations for all the physically motivated pairs of parameters presented in this figure exhibit opposite polarities, with $f_{\rm esc}$ increasing as one parameter increases and the other decreases, explaining why all observable ratios in Tab. 1 are effective diagnostics of $f_{\rm esc}$.

The values of $f_{\rm esc}$ and $\dot{N}_{\rm ion,esc}$ for the THESAN-ZOOM simulations are plotted in Fig. A4 against $M_{\rm UV}$ and M_*, analogously with Fig. 5 and the observational JWST catalogue. As with JWST, we use orthogonal distance regression to fit linear relationships between $\log_{10}(\dot{N}_{\rm ion,esc})$ and $M_{\rm UV}$ as well as $\log_{10}(M_*)$, again including a redshift dependence to account for the cosmic evolution of $\dot{N}_{\rm ion,esc}$; with the relations presented in their respective panels. Half the $16^{\rm th}-84^{\rm th}$ percentile range of the residuals of these relations, $\sigma_{\dot{N}}$, is also shown in both panels. Eq (7) states the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation found here in the main text. This relation, together with its intrinsic scatter of $\sigma_{\dot{N}} = 0.725$ dex, is used to derive the THESAN-ZOOM-based ionizing emissivity, $\dot{n}_{\rm ion}$, as explained in Section. 4.1.

APPENDIX B: OPTIMISING THE RANDOM FOREST

Here, we explain our processes for optimising the hyperparameters of our random forest (RF) models (see Section 3.2). For each model, we train RFs to target $f_{\rm esc}$ using all 14 of the relevant indicators given in Tab. 1. The Mean Absolute Error (MAE; in dex) reported for each instance of a hyperparameter in this analysis is the mean MAE calculated over 10 separate regressors with the same setup.

The first hyperparameter we address is the minimum number of samples on the final leaf, which dictates the minimum number of samples allowed to exist at an end node of a decision tree. This controls the size a decision tree in the RF can grow to. If too low, the regressor tends to overfit the training sample, however increasing it also decreases the accuracy of the sample (increasing MAE and MSE). We choose the value of the minimum number of samples on the final leaf to be 50, which is the smallest value that yields a difference in MAE between prediction on the testing and training samples of less than 5% (see Fig. B1).

We also optimise the number of features considered when looking for the best split at a node. If set too high, more parameters are evaluated at each split and the model tends to select the most important parameter repeatedly, limiting exploration of other parameters. This removes much of the random nature of the RF and with it the benefits mentioned in Section 3.2. We set the maximum features equal to the commonly used square root of the total number of features, for the 16 training features for $f_{\rm esc}$ this is a maximum of 4 in this work. Lowering beyond this causes regressor MAE to increase rapidly (see Fig. B2).

The final hyperparameter we address is the number of estimators, which is the number of decision trees in the random forest. A larger number of predictors unsurprisingly decreases errors, although the increase in accuracy falls off exponentially. Unfortunately, training time complexity increases linearly with the number of estimators, expressed as $O(n)$. We therefore conclude that 210 decision trees is an appropriate choice for our models; this is the lowest number of estimators that results in an MAE difference of less than 0.05% from a model trained with 1000 decision trees (see Fig. B3).

APPENDIX C: VALIDATION OF THE IONIZING EMISSIVITY INTEGRATIONS

In calculating the ionizing emissivities, $\dot{n}_{\rm ion}(z)$, of Fig. 6 with Eq. (1), we employ the Schechter form of the UV luminosity function (UVLF) (Schechter 1976):

$$\Phi(M_{\rm UV}, z) = 0.4\ln(10)\,\phi^* \cdot 10^{0.4\,(\alpha+1)\,\Delta M_{\rm UV}} \cdot \exp\left(-10^{0.4\,\Delta M_{\rm UV}}\right), \tag{C1}$$

where we have defined $\Delta M_{\rm UV} \equiv M^*_{\rm UV} - M_{\rm UV}$. In this work, the fit parameters $\phi^* \equiv \phi^*(z)$, $M^*_{\rm UV} \equiv M^*_{\rm UV}(z)$, and $\alpha \equiv \alpha(z)$ from Bouwens et al. (2021) are used for $z \in [1.5, 8.5]$ and those from Whitler et al. (2025) are used for $z \in [8.5, 16]$. Some sources have found the double power-law (DPL) form of the UVLF to better fit observations at high redshifts of $z \gtrsim 8$ (Bowler et al. 2020):

$$\Phi(M_{\rm UV}, z) = \frac{0.4\ln(10)\,\phi^*}{10^{-0.4\,(\alpha+1)\,\Delta M_{\rm UV}} + 10^{-0.4\,(\beta+1)\,\Delta M_{\rm UV}}}. \tag{C2}$$

In Fig. C1, we compare the THESAN-ZOOM-based $\dot{n}_{\rm ion}(z)$ curve obtained in Section 4.1, with an alternative THESAN-ZOOM-based curve derived employing the DPL form of the UVLF. The DPL fit parameters $\phi^* \equiv \phi^*(z)$, $M^*_{\rm UV} \equiv M^*_{\rm UV}(z)$, $\alpha \equiv \alpha(z)$, and $\beta \equiv \beta(z)$ are taken from (Bowler et al. 2020) for $z \in [3.5, 8.5]$ and from Whitler et al.

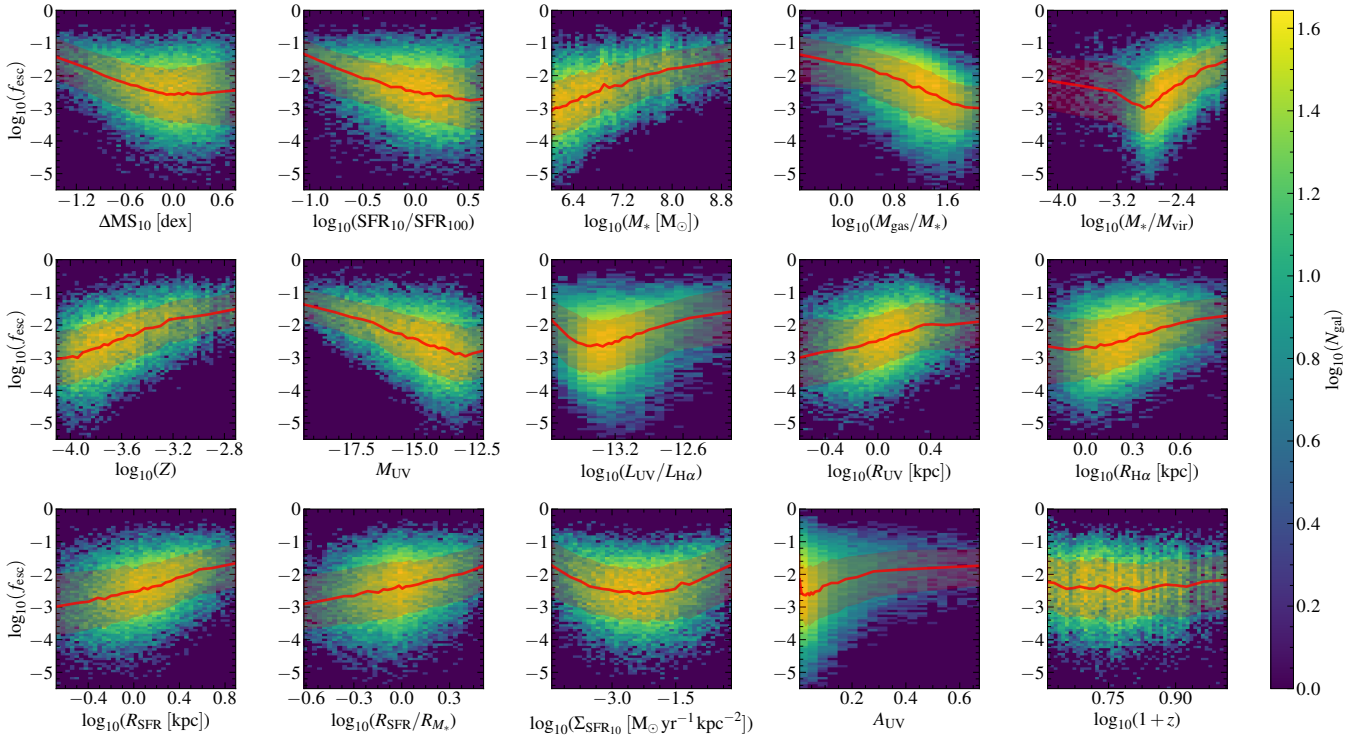


Figure A1. Ionising photon escape fraction ($f_{\rm esc}$) versus the 14 identified diagnostics listed in Tab. 1, as well as $1+z$, for the 35 512 galaxies selected from the THESAN-ZOOM simulations across $3 < z < 16$. The colour scale indicates the number of galaxies per bin. The red lines trace the median relations, with red shaded regions denoting the 16$^{\rm th}$–84$^{\rm th}$ percentiles. All the observables exhibit a clear correlation with $f_{\rm esc}$, demonstrating that they are all potentially good indicators of $f_{\rm esc}$.

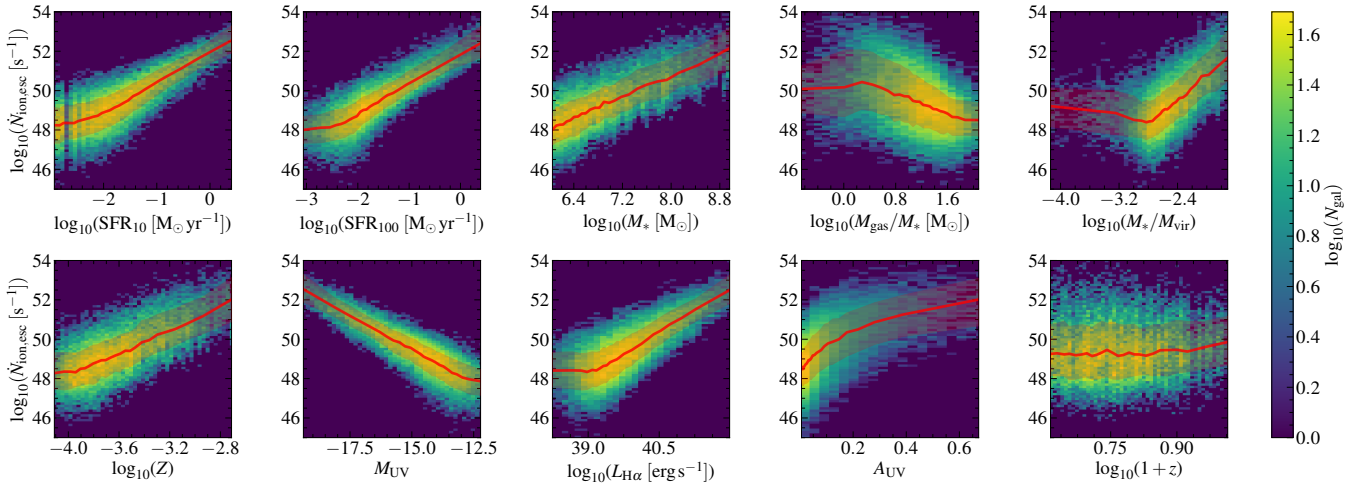


Figure A2. Ionising photon escape rate ($\dot{N}_{\rm ion,esc}$) versus the 9 identified diagnostics listed in Tab. 1, as well as $1+z$, for the 35 512 galaxies selected from the THESAN-ZOOM simulations simulations across $3 < z < 16$. The colour scale indicates the number of galaxies per bin. The red lines trace the median relations, with red shaded regions denoting the 16$^{\rm th}$–84$^{\rm th}$ percentiles. All the observables exhibit a clear correlation with $\dot{N}_{\rm ion,esc}$, demonstrating that they are all potentially good indicators of $\dot{N}_{\rm ion,esc}$.

(2025) for $z \in [8.5, 16]$. The emissivities derived from the Schechter and DPL UV luminosity functions are consistent for $z \leq 8$, the redshift range over which we find the bulk of the ionizing photon budget to be produced; variations in $\dot{n}_{\rm ion}(z)$ at earlier times therefore have little effect on the completion of reionization.. However, the DPL curve exhibits an unphysical discontinuous increase from $z \simeq 8$ to $z \simeq 10$, causing it to diverge from the Schechter-based result. We therefore conclude that the emissivity derived from the Schechter fit is more suitable for interpolation and for the analysis of the cosmic hydrogen ionization fraction $Q_{\rm HII}$ presented in Section 4.3.

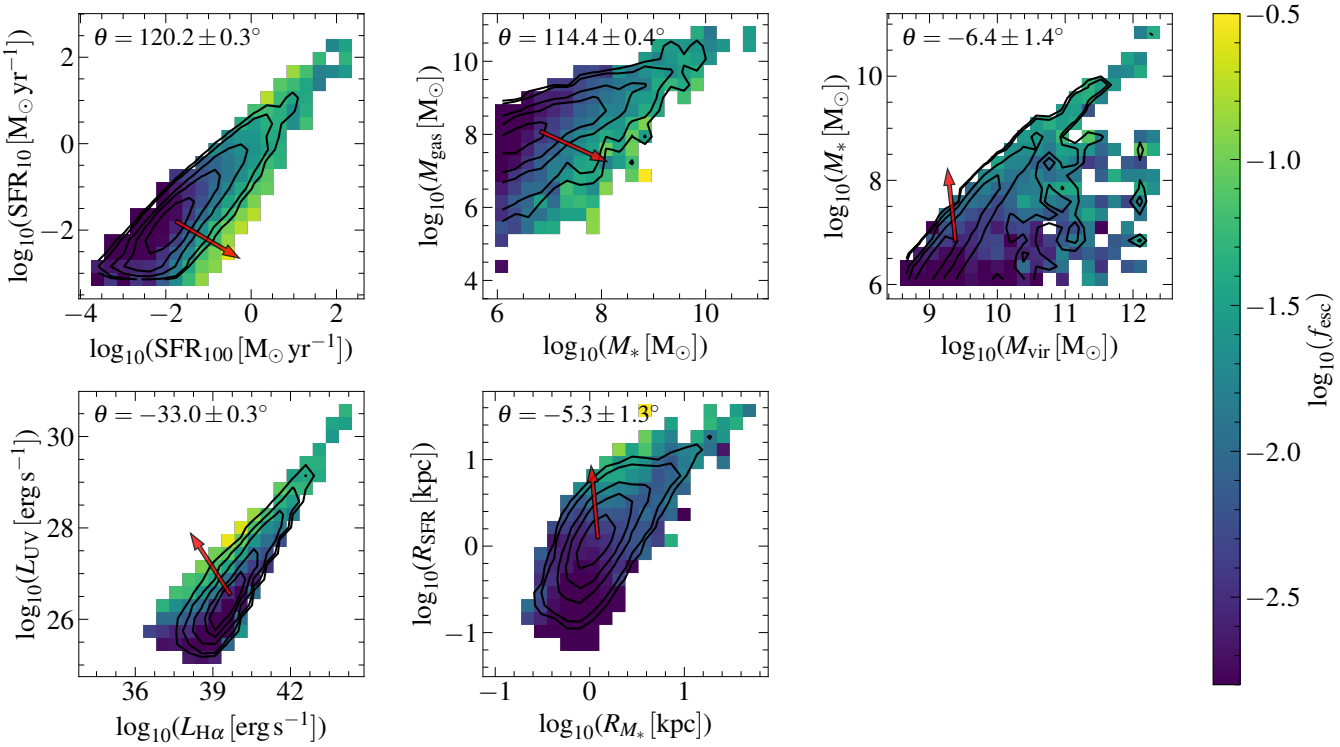


Figure A3. The pairs of observables that form the ionizing photon escape fraction ($f_{\rm esc}$) ratio indicators given in Tab. 1 for the THESAN-ZOOM galaxies are plotted against each other, colour-coded by the median $f_{\rm esc}$ in each bin. Only bins containing at least five galaxies are plotted. The contours indicate the density of galaxies in the plots, with the innermost contours encompassing 34.0% of the sample and the outermost contours encompassing 97.5%. The red arrows indicate the direction of largest positive colour gradient, with their angles, θ, defined clockwise from the positive y-direction. All θ lie in the regions $(-90, 0) \cup (90, 180)$, implying that the pairs of observables have correlations of opposite polarity with $f_{\rm esc}$. Consequently, $f_{\rm esc}$ increases as one observable increases and the other decreases, so their ratios are likely to be good indicators of $f_{\rm esc}$.

Eq. (1) can be reparametrized for $\dot{N}_{\rm ion,esc} - M_*$ relations by adopting galaxy number density functions expressed in terms of stellar mass, $\Phi(M_*, z)$, commonly referred to as stellar mass functions (SMFs):

$$\dot{n}_{\rm ion}(z) = \int_{M_{*,\rm min}}^{M_{*,\rm max}} \Phi(M_*, z) \left\langle \dot{N}_{\rm ion,esc} \,\middle|\, M_*, z \right\rangle {\rm d}M_*. \tag{C3}$$

To assess the viability of stellar mass functions, we again compare the THESAN-ZOOM-based $\dot{n}_{\rm ion}(z)$ of Fig. 6, with another alternative THESAN-ZOOM-based curve in Fig. C1. The latter being derived from the THESAN-ZOOM $\dot{N}_{\rm ion,esc} - M_*$ relation presented in the bottom-right panel of Fig. A4, employing the Schechter form of the stellar mass function (Schechter 1976):

$$\Phi(y_*, z) = \ln(10)\, \phi' \, 10^{(y_* - \mu)\,(\alpha+1)} \exp\left(-10^{(y_* - \mu)}\right), \tag{C4}$$

Where we have defined $y_* \equiv \log_{10}(M_*)$. We employ the fit parameters $\phi' \equiv \phi'(z)$, $\mu \equiv \log_{10}(M'_*(z))$, and $\alpha \equiv \alpha(z)$ from Weibel et al. (2024) for $z \in [3.5, 9.5]$. At high redshifts, the stellar mass derived $\dot{n}_{\rm ion}$ values are significantly larger than those derived from UV magnitudes, exhibit large uncertainties, appear less continuous, and do not decline sufficiently to avoid an overabundance of ionising photons, which results in unphysical $Q_{\rm HII}$ evolution. This behaviour likely reflects the well-known limitations of stellar mass functions at high redshifts (Madau & Dickinson 2014). This approach is therefore not well suited to the reionization analysis presented in Section 4. We therefore adopt Eq. (1) for the main results of this study and employ a Schechter parameterisation of the UV luminosity function.

As described in Section 4.1, to account for the scatter of $\sigma_{\dot{N}} = 0.725$ dex observed in the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation in THESAN-ZOOM when computing the ionizing emissivity, we multiply the integrand of Eq. (1) by the lognormal correction factor specified in Eq. (8). In doing this we have implicitly assumed that the scatter in this relation is constant across $M_{\rm UV}$. Fig. C2 shows the residuals of the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation for the THESAN-ZOOM galaxies as a function of $M_{\rm UV}$. In this figure, we see that the width of the 1σ percentile range varies only weakly with $M_{\rm UV}$ and exhibits no clear systematic trend that would justify parametrizing $\sigma_{\dot{N}}$ as a function of $M_{\rm UV}$. This supports our assumption of a constant scatter in the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation.

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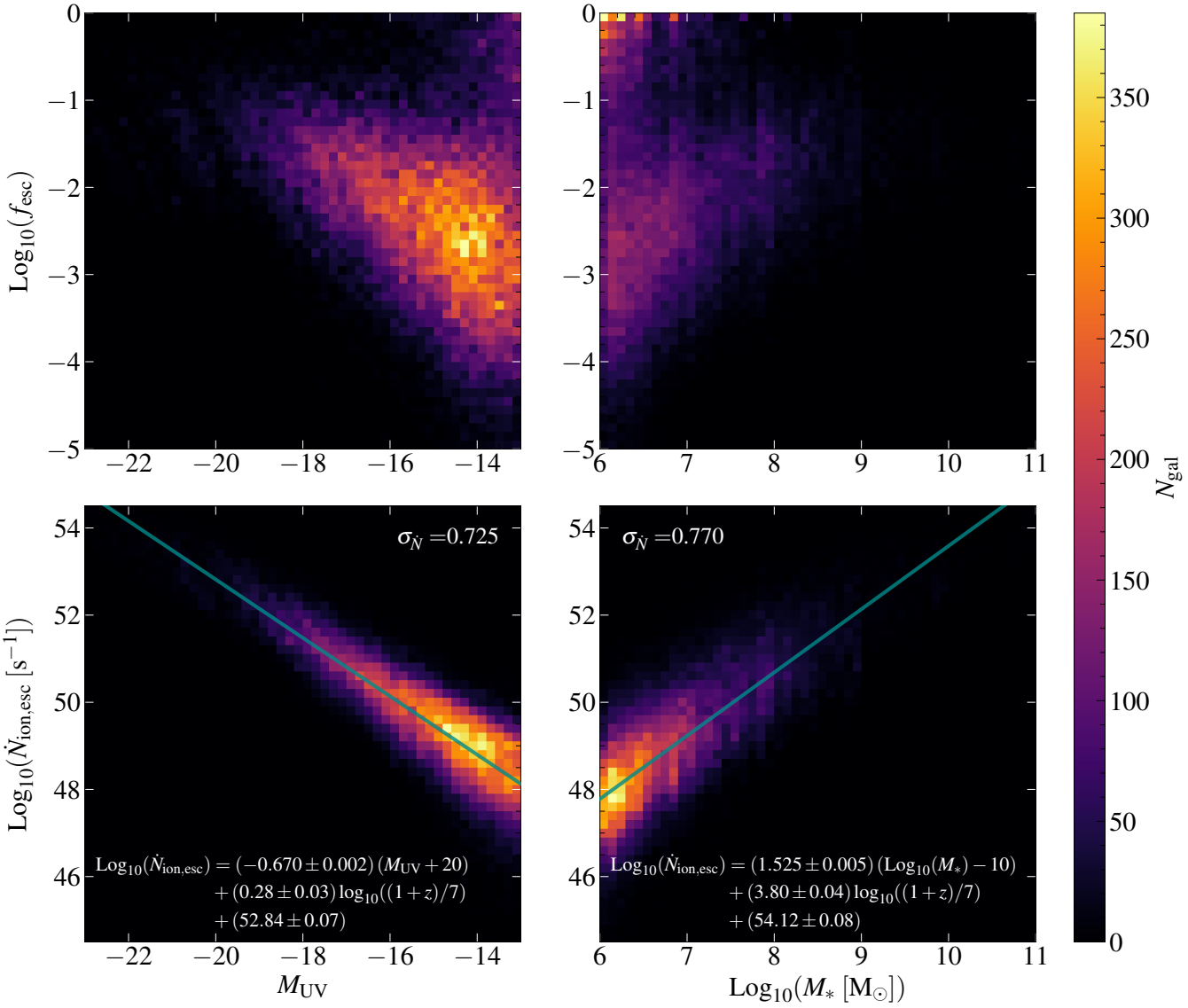


Figure A4. *Top left:* $f_{\rm esc}$ as a function of $M_{\rm UV}$ for galaxies from the THESAN-ZOOM simulations; *top right:* $f_{\rm esc}$ as a function of M_*; *bottom left:* $\dot{N}_{\rm ion,esc}$ as a function of $M_{\rm UV}$; *bottom right:* $\dot{N}_{\rm ion,esc}$ as a function of M_*. The colour scale indicates the number of galaxies per bin. Like with the predictions of the JWST observational sample in Fig. 5, $\log_{10}(\dot{N}_{\rm ion,esc})$ exhibits approximately linear behaviour with both $M_{\rm UV}$ and $\log_{10}(M_*)$. Therefore, in both lower panels, we present the orthogonal distance regression linear relations for $\log_{10}(\dot{N}_{\rm ion,esc})$, in which we also include a redshift dependence, and plot them in cyan. The scatter about these relations, $\sigma_{\dot{N}}$, computed as half the 16$^{\rm th}$ – 84$^{\rm th}$ percentile range of the residuals, is also shown in both panels. This scatter is slightly smaller for the $M_{\rm UV}$ relation, suggesting that $M_{\rm UV}$ may be the more important diagnostic of $\dot{N}_{\rm ion,esc}$, as we see in Fig. 3.

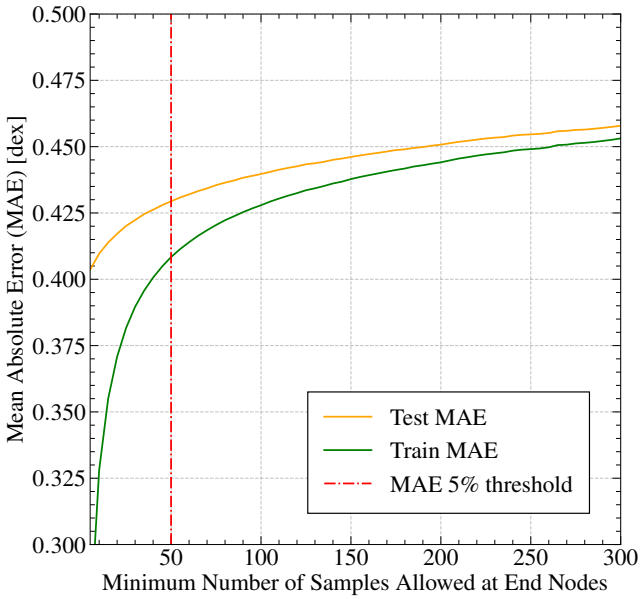


Figure B1. Mean Absolute Error (MAE; in dex) of $f_{\rm esc}$ prediction as a function of the minimum number of samples allowed on the end node of a random forest decision tree. The orange line shows the MAE of prediction on the test sample, while the green line shows the MAE of prediction on the training sample. Increasing this hyperparameter reduces overfitting, as evidenced by the decreasing difference in MAE between the two curves. However, larger values also reduce prediction accuracy. We therefore adopt the smallest value for which the difference in MAE between the test and training samples is less than 5%, which occurs at 50, as indicated by the dashed red line.

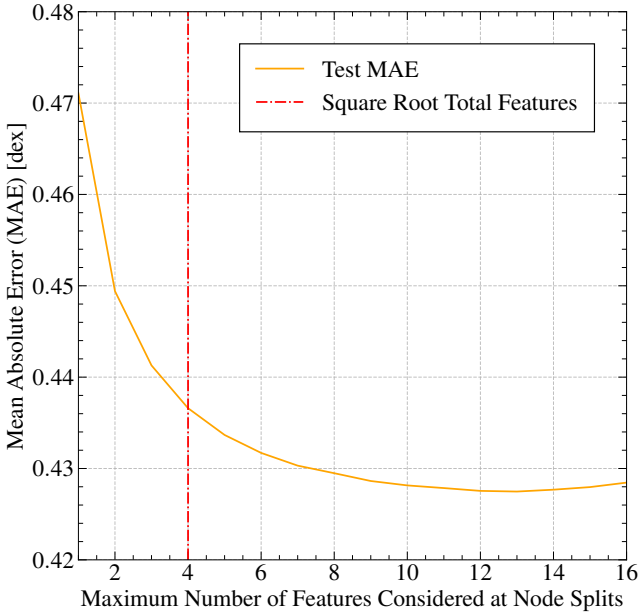


Figure B2. Mean Absolute Error (MAE; in dex) of $f_{\rm esc}$ prediction for the test sample as a function of the maximum number of features considered at each decision tree node split in the random forest. We set this hyperparameter equal to the square root of the total number of features (four for the 16 $f_{\rm esc}$ random forest features of this work), as indicated by the red dashed line.

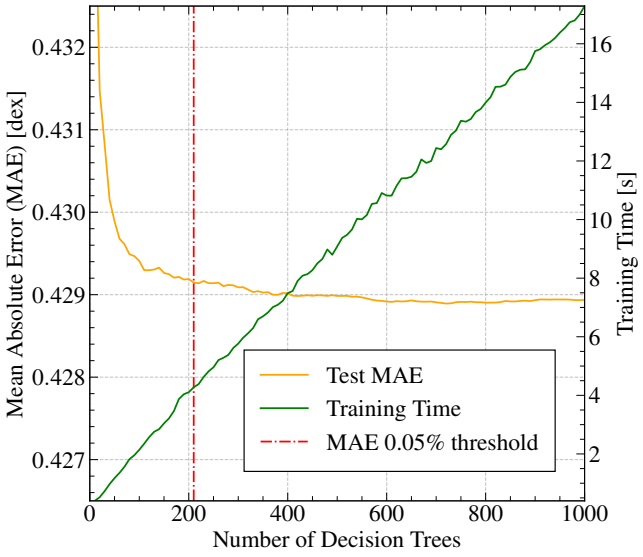


Figure B3. The Orange line shows the Mean Absolute Error (MAE; in dex) of $f_{\rm esc}$ prediction for the test sample as a function of the number of decision trees in the random forest. The green line illustrates the linear relationship between model training time and the number of trees. We set this hyperparameter equal to 210, the smallest number of estimators for which the MAE on prediction differs by less than 0.05% from that of a model trained with 1000 trees. This threshold is indicated by the red dashed line.

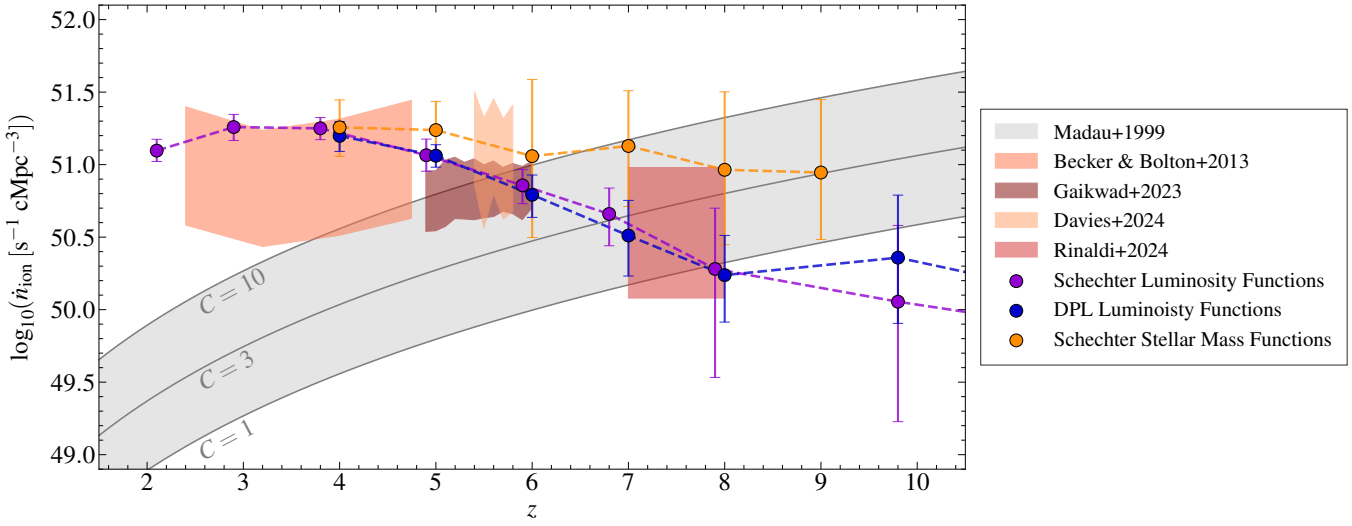


Figure C1. Similar to Fig. 6: comoving cosmic ionizing photon emissivity, $\dot{n}_{\rm ion}$, as a function of redshift, derived from the THESAN-ZOOM $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ and $\dot{N}_{\rm ion,esc} - M_*$ relations presented in Fig. A4. The violet emissivities have been integrated using the Schechter parametrization of the UV luminosity functions (see Eq. (C1)) from Bouwens et al. (2021) for $z \in [1.5, 8.5]$ and Whitler et al. (2025) for $z \in [8.5, 16]$. The blue emissivities have instead been obtained using the Double Power Law (DPL) parametrization of the UV luminosity functions (see Eq. (C2)) from Bowler et al. (2020) for $z \in [3.5, 8.5]$ and Whitler et al. (2025) for $z \in [8.5, 16]$. Finally, the orange emissivities have been derived using the Schechter parametrization of the stellar mass functions (see Eq. (C4)) from Weibel et al. (2024) for $z \in [3.5, 9.5]$. As in Fig. 6, the grey bands indicate the critical ionizing emissivity (Madau et al. 1999) for possible clumping factors $C \in [1, 10]$, and we include the same observational constraints (Becker & Bolton 2013; Gaikwad et al. 2023; Davies et al. 2024; Rinaldi et al. 2024).

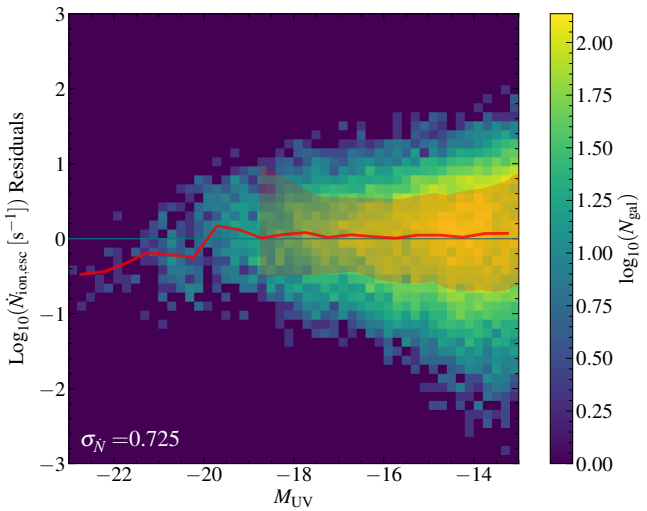


Figure C2. Residuals of ionizing photon escape rate ($\dot{N}_{\rm ion,esc}$) as a function of rest-frame UV absolute magnitude ($M_{\rm UV}$), relative to the redshift dependent $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation given by Eq. (7) for THESAN-ZOOM galaxies. The colour scale indicates the number of galaxies per bin. The red shaded region denotes the $16^{\rm th} - 84^{\rm th}$ percentile range, calculated in bins containing equal numbers of galaxies, while the red line denotes the median calculated in bins of equal width in $M_{\rm UV}$. The former provides a robust representation of the intrinsic scatter, while the latter allows the median relation to be extended to the sparsely populated bright end. We see the width of the range varies only weakly with $M_{\rm UV}$ and shows no clear systematic trend, implying that the assumption of constant scatter in the $\dot{N}_{\rm ion,esc} - M_{\rm UV}$ relation is well motivated. We take this constant scatter to be half of the $16^{\rm th} - 84^{\rm th}$ percentile range over all the residuals, and report its value in the lower left of this figure.